Event cameras



# ESVO2: Direct Visual-Inertial Odometry With Stereo Event Cameras

Junkai Niu, Sheng Zhong, Xiuyuan Lu, Shaojie Shen, Guillermo Gallego, and Yi Zhou, *Member, IEEE*

***Abstract*—Event-based visual odometry is a specific branch of visual simultaneous localization and mapping (SLAM) techniques, which aims at solving tracking and mapping subproblems (typically in parallel), by exploiting the special working principles of neuromorphic (i.e., event-based) cameras. Due to the motion-dependent nature of event data, explicit data association (i.e., feature matching) under large-baseline viewpoint changes is difficult to establish, making direct methods a more rational choice. However, state-of-the-art direct methods are limited by the high computational complexity of the mapping subproblem and the degeneracy of camera pose tracking in certain degrees of freedom (DoF) in rotation. In this article, we tackle these issues by building an event-based stereo visual-inertial odometry system, which is built upon a direct pipeline known as event-based stereo visual odometry (ESVO). Specifically, to speed up the mapping operation, we propose an efficient strategy for sampling contour points according to the local dynamics of events. The mapping performance is also improved in terms of structure completeness and local smoothness by merging the temporal stereo and static stereo results. To circumvent the degeneracy of camera pose tracking in recovering the pitch and yaw components of general 6-DoF motion, we introduce IMU measurements as motion priors via preintegration. To this end, a compact back-end is proposed for continuously updating the IMU bias and predicting the linear velocity, enabling an accurate motion prediction for camera pose tracking. The resulting system scales well with modern high-resolution event cameras and leads to better global positioning accuracy in large-scale outdoor environments. Extensive evaluations on five publicly available datasets featuring different resolutions and scenarios justify the superior performance of the proposed system against five state-of-the-art methods. Compared to ESVO, our new pipeline significantly reduces the camera pose tracking error by 40%–80% and 20%–80% in terms of absolute trajectory error and relative pose error, respectively; at the same time, the mapping efficiency is improved by a factor of five. We release our pipeline as an open-source software for future research in this field.**

***Index Terms*—Event-based cameras, simultaneous localization and mapping, visual odometry, visual-inertial fusion.**

## I. Introduction

DRAWING inspiration from biological systems, event cameras represent a revolutionary advancement in sensor technology. These sensors transmit pixelwise intensity changes, called "events," asynchronously as they occur [3], [4], [5]. Consequently, they do not produce grayscale images, nor do they adhere to an external clock of fixed capture rate typical of traditional cameras. This asynchronous and differential principle of operation minimizes temporal redundancy, thereby reducing power consumption and bandwidth requirements. Endowed with high temporal resolution and high dynamic range (HDR) capabilities [4], [5], event cameras are qualified to deal with challenging scenarios that are inaccessible to traditional cameras, such as high-speed motion and/or HDR illumination tracking [6], [7], [8], [9], [10], [11], [12], [13], control [14], [15], [16], [17], [18], [19], [20], [21], image synthesis [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], and simultaneous localization and mapping (SLAM) [32], [33], [34], [35], [36], [37], [38], [39], [40].

Although the potential of event cameras is promising, their unconventional output is incompatible with existing modules in visual odometry (VO) / SLAM systems designed for traditional cameras. Event-based VO, on the one hand, inherits the theoretical formulation and parallel design from modern VO solutions. On the other hand, the distinctiveness of event-based methods is witnessed in several aspects, including data processing and representation. Early works using a monocular event camera (i.e., [33] and [35]) require a very gentle motion (typically a small-loopy behavior) for the initialization of a local 3-D map, based on which the camera pose can be tracked using a 3-D-2-D registration pipeline. The dependence on dedicated hardware (e.g., a GPU) [33] and the inefficient map expansion [33], [35] restrict their applications on mobile platforms equipped with a limited computation resource and energy supply. More recently, Zhou et al. [1] proposed an event-based stereo visual odometry (ESVO) pipeline that achieves better accuracy and demonstrates a real-time performance on data collected by a moving agent, such as a flying drone or a ground vehicle. However, this method does not scale well with the streaming rate of event cameras. Real-time performance is only witnessed on event cameras with a small spatial resolution (Q-VGA), e.g., a DAVIS346 (dynamic

Received 13 January 2025; accepted 26 February 2025. Date of publication 5 March 2025; date of current version 27 March 2025. This work was supported by the National Key Research and Development Project of China under Grant 2023YFB4706600. This article was recommended for publication by Guest Associate Editor G. Cioffi and Editor J. Civera upon evaluation of the reviewers' comments. *(Junkai Niu and Sheng Zhong contributed equally to this work.) (Corresponding author: Yi Zhou.)*

Junkai Niu, Sheng Zhong, and Yi Zhou are with the Neuromorphic Automation and Intelligence Lab (NAIL) at School of Robotics, Hunan University, Changsha 410012, China (e-mail: eeyzhou@hnu.edu.cn).

Xiuyuan Lu and Shaojie Shen are with the Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology, Hong Kong, China.

Guillermo Gallego is with the TU Berlin, the Science of Intelligence Excellence Cluster, the Robotics Institute Germany and the Einstein Center Digital Future, Berlin 10587, Germany.

Data is available online at https://youtu.be/gmAU32Oeiv8; https://github.com/NAIL-HNU/ESVO2.git.

This article has supplementary downloadable material available at https://doi.org/10.1109/TRO.2025.3548523, provided by the authors.

Digital Object Identifier 10.1109/TRO.2025.3548523

1941-0468 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

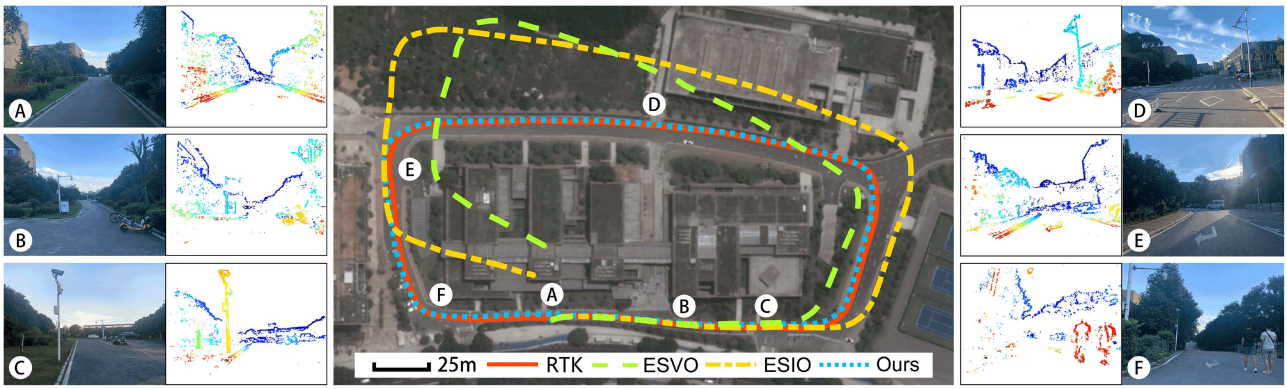



Fig. 1. Proposed event-based stereo visual-inertial odometry system (ESVO2) operates in real time on large-scale VGA-resolution sequences like *hnu_campus* utilizing a standard CPU. Middle: The ground truth of the trajectory is provided by real-time kinematic (RTK) positioning (red) and is aligned with Google Maps. Our method achieves the best performance in terms of global trajectory consistency, while significant drift is witnessed in the results of ESVO [1] (green) and ESIO [2] (yellow). Left and right: Depth maps reconstructed by our system and corresponding scene images (for visualization purposes only). The locations where visual measurements are captured are associated to the map using letters with a circle around them.

and active-pixel vision sensor) [41]. Also, the failure of tracking in the presence of a sudden variation in angular velocity casts the necessity of introducing an inertial measurement unit (IMU) into the system.

The goal of this article is to lift the above-mentioned limitations of the original ESVO framework. To this end, we extend ESVO and present a method for visual-inertial odometry (VIO) with a stereo event camera and an IMU. As demonstrated in Fig. 1, the proposed system achieves better mapping and tracking performance than the event-based state of the art (e.g., [1] and [2]) in terms of accuracy and efficiency due to the following efforts.

Contributions:

1) A novel image-like representation of events, which utilizes their local dynamics for an adaptive accumulation (AA). It is used for efficiently determining pixel locations associated with instantaneous edges (see Section IV-A);
2) An improved solution to the mapping subproblem by leveraging both the temporal-stereo and static-stereo configurations and, in addition, a fast block-matching scheme in the static-stereo operation (see Section IV-B);
3) Incorporating IMU preintegration as an initial value for event camera pose estimation (see Section IV-C), and further constructing a tightly-coupled visual-inertial backend that keeps suppressing the drift in the bias of the gyroscope and the accelerometer (see Section V);
4) An extensive experimental evaluation on five publicly available datasets and comparing against five methods, reporting state-of-the-art performance in accuracy and efficiency (see Section VI). It also shows that the system scales well with event cameras of different spatial resolutions and is able to run stably in real time on a standard CPU with data of up to VGA resolution. Besides, our dataset and implementation are open-sourced.

This work has evolved from our ICRA paper [42], and the technical improvements are clarified at the end of Section II.

*Outline:* The rest of this article is organized as follows. First, a literature review of relevant work is provided in Section II. Second, the proposed framework is then overviewed from a systematic perspective (see Section III). Then, we discuss our method by detailing each item listed in the contribution (see Sections IV and V). Experimental results are provided in Section VI. Finally, Section VII concludes this article.

## II. Related Work

Like its standard-vision counterparts, event-based VO/SLAM also aims to solve simultaneously the mapping and tracking subproblems in a recursive manner. The main challenge therein is to answer the following question: *How to process raw events in a way that photometric and geometric constraints can be established to infer the camera pose and scene depth information*? To this end, several data processing pipelines that utilize different sensor configurations and exploit various data representations have been proposed, as summarized in Table I. From the perspective of how event data are processed, existing methods (also including event-based VIO) can be mainly divided into two categories (see 2nd column of Table I): indirect methods and direct methods.

*Indirect Methods:* To build on top of existing indirect methods (e.g., [43] and [44]) using standard cameras, researchers have resorted to developing hand-crafted features from event data, such as event corners [45], [46], [47], [48], which are typically adapted from the original Harris [49] and FAST [50] methods. In addition, strategies for tracking event corners are presented by [47], [51]. Such event features enable straightforward application of mature geometric tools, such as epipolar geometry [52], [53] and perspective-*n*-point methods [54], [55]. Despite the success of these feature-based solutions (see, e.g., [56] and [57]), event features are not as theoretically robust as their standard-vision counterparts. This is due to the motion-dependent nature of event data, which sometimes leads to incomplete observation of junctions. Consequently, feature matching can easily fail in a sudden variation of the event camera's motion. Another mainstream strategy for feature detection and tracking is inspired by the motion compensation method [58], a unified pipeline for event-based model fitting. This strategy is widely witnessed in event-based VIO pipelines [2], [36], [59], which typically build features from

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I
LITERATURE REVIEW ON REPRESENTATIVE EVENT-BASED VO/VIO SYSTEMS

| Method | (In)Direct | Sensors | Event representation | Real-time performance | Evaluation datasets |
|---|---|---|---|---|---|
| Kim *et al.* [33] | Direct | Monocular | Raw events | (128 × 128 px) on GPU | – |
| Rebecq *et al.* [35] | Direct | Monocular | Raw events and Naive Accum. | (240 × 180 px) on CPU | ECD |
| Zhu *et al.* [59] | Indirect | Monocular + IMU | Raw events | ✗ | ECD |
| Rebecq *et al.* [36] | Indirect | Monocular + IMU | Motion-compensated Event Images | (240 × 180 px) on CPU | – |
| ESVO [1] | Direct | Stereo | TSs | (346 × 260 px) on CPU | RPG, MVSEC, HKUST |
| Hadviger *et al.* [56] | Indirect | Stereo | TSs | (346 × 260 px) on CPU | MVSEC, DSEC |
| ESIO [2] | Indirect | Stereo + IMU | TSs | (346 × 260 px) on CPU | HKU, MVSEC, VECtor |
| Wang *et al.* [57] | Indirect | Stereo | TSs and Naive Accum. | ✗ | MVSEC |
| ESVIO [63] | Direct | Stereo + IMU | TSs | (346 × 260 px) on CPU | RPG, MVSEC, ESIM, DSEC |
| Elmoudni *et al.* [64] | Direct | Stereo | Raw events and TSs | ✗ | DSEC |
| Shiba *et al.* [65] | Direct | Monocular / Stereo | Raw events | ✗ | RPG, DSEC |
| ES-PTAM [66] | Direct | Stereo | Raw events and Naive Accum. | (346 × 260 px) on CPU | RPG, MVSEC, DSEC, EVIMO2, TUM-VIE |
| Niu *et al.* [42] | Direct | Stereo + IMU | TSs and Adaptive Accum. | (346 × 260 px) on CPU | RPG, DSEC |
| **This work** | Direct | Stereo + IMU | TSs and Adaptive Accum. | (640 × 480 px) on CPU | RPG, MVSEC, DSEC, VECtor, TUM-VIE |

motion-compensated event sets [8] or event histograms [60] and further fuse with inertial measurements via either a Kalman filter [61] or keyframe-based nonlinear optimization [62].

*Direct Methods:* Unlike feature-based methods, direct methods refer to those that directly process either events or raw pixel information via some intermediate representation of event data (see 4th column of Table I). Based on the constant-brightness assumption in logarithmic scale, Kim et al. [33] proposed the first direct method, consisting of three interleaved probabilistic filters to solve the subproblems of mapping, camera pose tracking and, in addition, recovering the intensity information. To justify that recovering the intensity information is not needed and that a GPU is not required, Rebecq et al. [35] proposed a geometric approach. The mapping module determined the 3-D location of scene structures by searching for local maxima of ray densities (represented by a disparity space image – DSI), and the camera pose was estimated by 3-D-2-D registration, aligning the local 3-D map to the 2-D event locations on the image plane. These two pioneering works are, however, limited by the requirements of gentle motion in the initialization and slow expansion of the local map.

To overcome these limitations, Zhou et al. [1] presented the first event-based VO pipeline (ESVO) using a stereo event camera. The method exploits spatio-temporal consistency of the events across the image planes of the cameras to solve both localization and mapping subproblems of VO. Nevertheless, ESVO does not achieve real-time performance for camera spatial resolutions of $640 \times 480$ pixels (VGA) or higher. This is mainly due to the large number of redundant operations in the mapping module, which are originally caused by the way contour points are determined. Besides, we observe that ESVO's tracking module sometimes cannot fully recover the yaw and pitch components in general 6-DoF motion, which degrades the accuracy of the recovered trajectory. To resolve these issues, we recently proposed an IMU-aided version [42] of ESVO. By employing an efficient strategy for sampling contour points, it scales better with modern event cameras of VGA resolution. The degeneracy issue of camera pose tracking is circumvented by introducing as a prior the gyroscope measurements via preintegration. We also noticed the recent emergence of a direct method using a stereo event camera and an IMU [63]. It is built on top of ESVO, with an additional back-end that refines camera poses using IMU preintegration constraints. Although it outperforms ESVO in terms of trajectory accuracy, the core limitations of ESVO mentioned above are not well resolved.

Some other follow-up works of ESVO, such as [64], specifically focus on improving the accuracy of the mapping subproblem using stereo-fused ray densities [66], [67]. The resulting semidense depth maps are sharper and more accurate, however, real-time performance on VGA spatial resolution cameras has never been reported. Finally, Shiba et al. [65] recently extended the contrast maximization framework [58], [68] to estimate optical flow and ego-motion from event data. With special consideration on the space-time properties of event data, it can handle overfitting in the event-alignment problem, thus leading to better performance in geometric-model-fitting tasks, including ego-motion estimation. Nevertheless, this theoretical work does not demonstrate real-time performance.

The system proposed in this article is built on top of our recent ICRA work [42] (also in Table I) with the following extensions and improvements:

1) The AA scheme originally proposed in [42] is nontrivially improved here. Compared to the original design, we introduce the idea of event-dynamic monitoring [69] into the termination criteria of event accumulation. This modification makes AA responsive to event rate variations without manually adjusting parameters. To achieve real-time performance, we also derive an approximate formula to calculate the converged event activity;
2) Inheriting the combination of temporal-stereo and static-stereo configurations from [42], we significantly reduce the computational cost of static-stereo operation in this article by proposing a fast block-matching scheme, which ensures the mapping module to run stably at 20 Hz on event data of VGA spatial resolution;
3) Using IMU preintegration as motion priors was naively proposed in [42], however, the IMU bias was not effectively suppressed. To solve this problem while ensuring real-time performance, we create a compact and efficient back-end, which considers only linear velocity and IMU biases as optimizing variables. Using the back-end's results as the initial values for the spatio-temporal registration leads to more efficient and accurate performance of camera pose tracking. According to our investigation on various back-end designs (see, e.g., [63]), our model strikes a good balance between accuracy and efficiency;

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

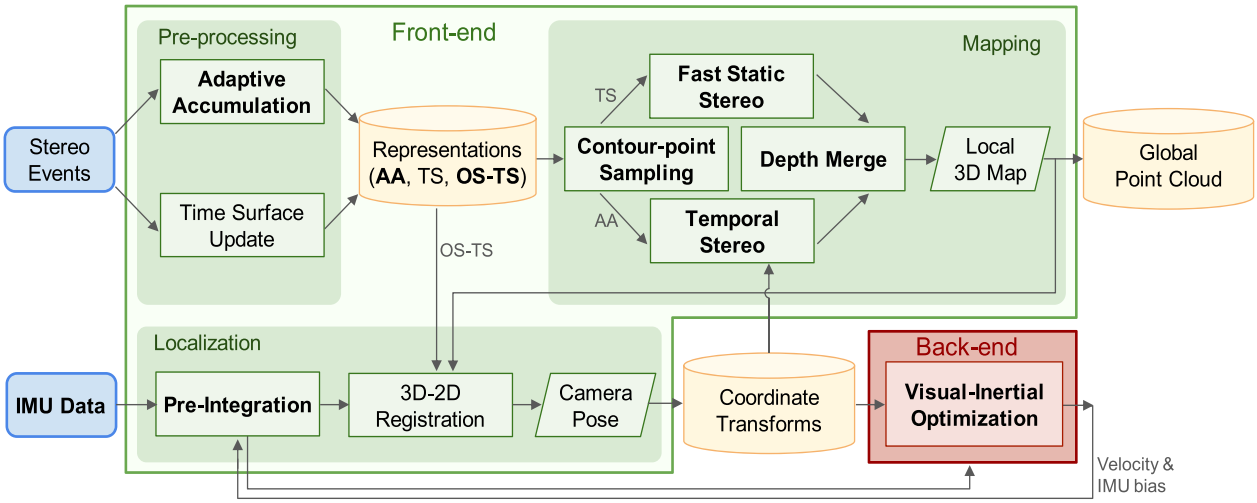



Fig. 2. *Flowchart of the proposed system (ESVO2).* New functions added to the original ESVO framework are highlighted in bold. Each of the four main modules (preprocessing, localization, mapping, and back-end) is executed independently and occupies at least one thread. The outputs (rhomboidal shapes) consist of the "local 3-D map" and the "camera pose," which can be stored to produce coordinate transforms and a global point cloud (in yellow).

4) The proposed pipeline is more extensively evaluated. Compared to [42], ESVO2 reduces the trajectory error by approximately 50% across all datasets and improves the computational efficiency of the mapping subproblem by more than a factor of two.

## III. System Overview

Using the original ESVO [1] pipeline as a backbone, we extend it into a stereo VIO system. Given as input the raw events from a calibrated stereo event camera and the inertial measurements from an IMU, the goal is identical to that of ESVO, namely, estimating the stereo event camera's pose in real time and recovering a semidense depth map and, furthermore, aiming at a better performance in terms of accuracy and efficiency. Fig. 2 displays the flowchart of the system, with the newly added functions highlighted in bold. The entire system can be primarily divided into four modules: preprocessing of event data, mapping, camera-pose estimation, and a back-end that keeps updating the IMU bias and the camera's linear velocity. The first three modules constitute what is referred to as the front-end.

To keep this section self-contained, let us briefly introduce the functions in each module, with a particular emphasis on new components added to the original ESVO pipeline.

First of all, the preprocessing module generates two event representations, including the time surface (TS) map (refer to details in [1, Sec. III-A]) and our novel representation called the AA map of events (see Section IV-A). Both of them can be refreshed in either an asynchronous manner (e.g., at the occurrence of a certain number of events) or at a fixed rate (e.g., 100 Hz in our implementation) and are stored separately in a database for access by other modules.

Second, IMU preintegration is used as the motion prior to initialize the camera pose estimator, which is formulated as a 3-D-2-D spatio-temporal registration problem (alignment of the 3-D local map projected onto the time-surface domain). The resulting pose estimates are stored in a database of coordinate transforms (e.g., TF in ROS [70]).

Third, the mapping module takes the contour points sampled on the AA maps, TSs, and camera pose estimates to keep updating a semidense local depth map. Specifically, two kinds of stereo methods are deployed to handle points with different observability properties on the spatio-temporal profile of events. The resulting depth estimates are merged to refresh the local map, which is constantly pushed into a database of global point cloud for potential visualization. Finally, the back-end realizes a sliding-window optimization that keeps updating the estimates of IMU biases and camera velocity at a fixed rate.

The bootstrap of our system creates an initial local map and initializes the unknowns in the back-end (i.e., the IMU biases and camera velocity). The former operation is identical to that in ESVO. A loosely-coupled method is performed in the latter operation, which determines the back-end parameters by aligning two trajectories obtained independently from events and IMU measurements. Different from what has been proposed in the initialization of [63], we find that the refinement of scale is unnecessary as long as the extrinsic parameters are sufficiently accurate.

## IV. Front-End

We detail our front-end in this section. First, we introduce a novel image-like representation of event data that facilitates efficient sampling of contour points (see Section IV-A). Second, we discuss our modifications to ESVO's mapping that lead to improvements in terms of structure completeness and local smoothness (see Section IV-B). Then, we demonstrate how to further enhance the camera pose estimation by incorporating the IMU preintegration as motion priors and employing offset-free time surfaces in the spatio-temporal registration (see Section IV-C).

The front-end interacts with a compact and efficient back-end (see Section V) that helps suppress drift in the estimated IMU biases, which is key to globally consistent trajectory estimates.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

### A. Preprocessing: AA of Events

The computational efficiency of the mapping method in ESVO is limited by several aspects, and one of them is the way that the edge-pixel locations are determined. In ESVO, an edge map in a virtual reference frame is created by applying motion compensation to events occurring within a short time interval (e.g., 10 ms). This operation will become computationally expensive as a preprocessing step when the streaming rate of events exceeds a certain range.[1] Besides, we observe that the extracted events are typically concentrated in certain regions with large optical flow. To alleviate such an uneven distribution of contour points, the method has to sample a large number of points, which is redundant and becomes a computational burden to mapping. Therefore, a more efficient way is needed to determine contour points.

The synthesized event map obtained by a naive accumulation of events in [35] can be used as an approximate edge map. However, this approximation can become considerably inaccurate (i.e., blurred edges or invisible edges) when the depth varies significantly in the scene. This is because a global threshold for event accumulation cannot handle diverse local dynamics of events. To address this, we propose AA, a novel representation of event data, that can control the amount of events to be accumulated according to the local event dynamics. In [69], the event dynamic is represented by event activity $a(t, s)$, which is defined as the number of events within a given time interval $(s, t]$. A temporal decay strategy is applied to the event activity, which is updated every time an event arrives. The event activity at the arrival of the $i_{\text{th}}$ event can be represented as follows:

$$a_i = \beta_i a_{i-1} + 1 \tag{1}$$

where $\beta_i$ is the decay rate that varies with event activity $a_{i-1}$ and is determined by the time interval $(t_i - t_{i-1})$ between two consecutive events as follows:

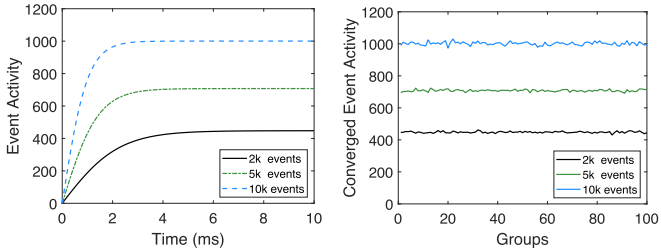
$$\beta_i = \frac{1}{1 + a_{i-1}(t_i - t_{i-1})}. \tag{2}$$

The method in [69] utilizes the event activity within a time interval to represent the global event dynamics, controlling the decay rate depending on the event activity to generate adaptive-decay time surfaces. Inspired by [69], our method monitors the event activity to determine the termination time of event accumulation. Intuitively, a greater event activity corresponds to higher event dynamics, indicating a shorter duration for event accumulation. We also observe that the event dynamics remain relatively stable within a short time interval (e.g., 10 ms), which leads to event activities converging toward a constant value incrementally, as shown in Fig. 3(a). Assuming the events are generated at equal time intervals, the converged event activity can be roughly calculated as follows:

$$\lim_{t_i \to t} a_i = (1 + \sqrt{1 + 4/b})/2 \approx 1/\sqrt{b},\ t_i \in (s, t] \tag{3}$$

where $b$ denotes the length of the time interval in seconds. As seen in Fig. 3(b), this estimation of converged event activity also holds approximately when the time interval $b$ is not constant. Therefore, we terminate event accumulation when the event activity converges to obtain a sufficiently sharp image. To handle the spatial variation of local event dynamics, we divide the image plane into small regions and accumulate events independently according to the local event activity.

[1]The event streaming rate is in proportion to several aspects, including the scene dynamics, scene texture, and spatial resolution of event cameras.




(a) Convergence behavior of event activity. (b) Irrelevance to temporal distribution of events.

Fig. 3. *Convergence of event activities.* (a) Illustrates the converging process of event activities with three different event dynamics (2 k, 5 k, and 10 k events per 10 ms, respectively). (b) Reports that the value of event activities at convergence is roughly independent of whether events are triggered uniformly in the temporal domain, as justified by 100 groups of independent simulations.

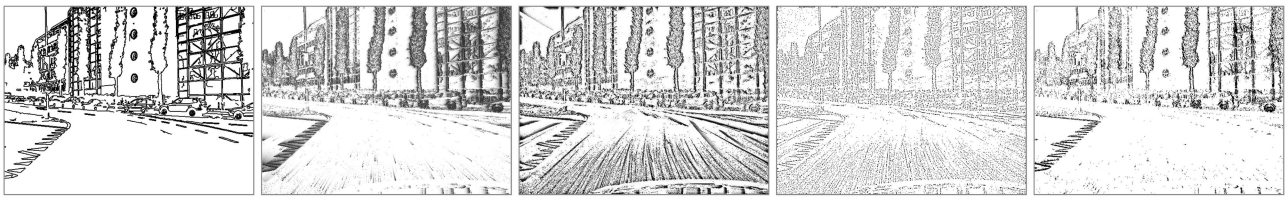
**Algorithm 1:** Adaptive Accumulation of Events.

**Input:** All involved events $\{e_k \doteq (x_k, y_k, t_k, p_k)\}_{k=1}^{N_e}$, and the amount of regions $N$.
**Output:** Adaptive accumulation map $\mathbf{A}(x, y)$.

```
Initialize A(x, y) with zero values.
Divide A(x, y) into N regions and assign each one a boolean flag F_j = False and two variables a_j and a_j^conv for storing the value of event activities.
for j = 1, ..., N do
    Estimate the average time interval b^j within the j_th region and calculate a_j^conv using (3).
    for i = 1, ..., N_e^j do
        if F_j == True then
            Break.
        end if
        Update a_j using (1) and (2).
        A(x_i, y_i)++.
        if |a_j − a_j^conv| / a_j^conv < 0.02 then
            F_j = True.
        end if
    end for
end for
```

The generation of an AA map is summarized in Algorithm 1. Specifically, we first divide the image plane evenly into $N$ small regions and the accumulation of events is carried out in each region independently. Events are sequentially accumulated in a region of an event count image until the event activity converges to the steady state value estimated by (3). The pixel value in an AA map represents the number of events accumulated at this pixel. Pixels with higher values are more likely to be associated with contour points.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e)

Fig. 4. "Event maps" generated using different event data representations, including TS [71], SILC [72], TOS [73], and the proposed AA map. The data used are from the *dsec_city_04a* sequence, which is recorded using a Prophesee Gen3.1 (VGA) event camera. The leftmost edge map is extracted from an intensity image and used for visualization only. (a) Edge map (from image). (b) TS [71]. (c) SILC [72]. (d) TOS [73]. (e) AA.

TABLE II
MULTICOMPONENT SSIM (3-SSIM) OF VARIOUS EVENT MAPS COMPARED TO THE EDGE MAP OBTAINED FROM THE CORRESPONDING INTENSITY IMAGE

| Event map | TS [75] | SILC [72] | TOS [73] | AA |
|---|---|---|---|---|
| 3-SSIM | 0.1801 | 0.1369 | 0.15203 | **0.3387** |

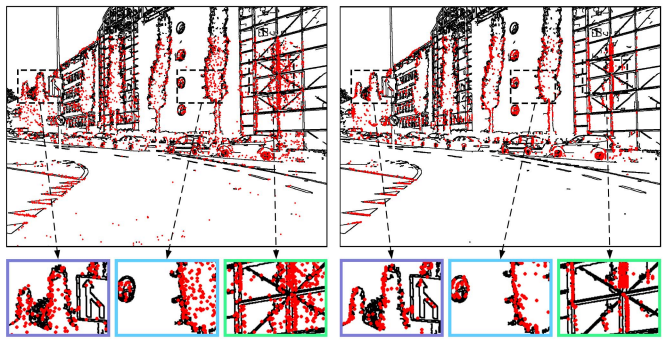
The data used for the 3-SSIM computation are identical to those used in Fig. 5.

To assess the effectiveness of the proposed AA method, we compare it against three image-like representations, including time surfaces [71] and two speed-invariant ones, namely, *speed invariant learned corners* (SILC) [72] and *threshold-ordinal surface* (TOS) [73]. As illustrated in Fig. 4, the AA map preserves relatively complete edges with fewest redundant points while keeping the highest true-edge-point ratio (TEPR).[2] This superior performance can also be justified quantitatively by comparing each representation against the edge map obtained from the corresponding grayscale image. As shown in Table II, the proposed AA representation leads to the highest score using the multicomponent structure similarity index measure (3-SSIM) [74] as evaluation metric.

Contour points are further sampled from each region independently. In general, the higher the AA pixel value in each region, the greater the probability that the pixel will be selected. We compare the sampling result to the contour points obtained by ESVO. As shown in Fig. 5, our sampled pixels better capture the edge structures, indicating that our sampling method can cover the edge patterns in the environment using fewer redundant input points.

### B. Mapping: Depth Estimation

The mapping module aims at recovering depth information and maintaining a local 3-D map. In the original ESVO pipeline, these are achieved by constantly propagating recent depth estimates of sampled events to a virtual reference frame, on which overlapped estimates are further fused based on their probabilistic characteristics. As for estimating the depth of each individual event, a two-step method (i.e., a block matching operation followed by a nonlinear refinement) is proposed. More specifically, a triangulation operation is performed given the resulting disparity from the block matching, and the nonlinear refinement operates in a forward-projection manner (i.e., from the inverse depth domain to 2-D event measurements). In general, the applied static stereo configuration in ESVO leads to unreliable depth estimation of structures parallel to the stereo baseline. Moreover, the two-step method for depth estimation does not scale well with the number of individual queries. To circumvent these two problems, we propose two modifications: 1) An additional temporal-stereo matching method and 2) a faster design of block matching. We detail each modification in the following.



(a) (b)

Fig. 5. *Point sampling.* (a) and (b) compare the results of sampling 4000 points by ESVO and our method, respectively. The background is an edge map obtained by processing the grayscale image with an edge detection algorithm. The values in parenthesis denote the proportion of contour points (black) covered by the sampled points (red). (a) ESVO (TEPR: 54.6%). (b) Ours (TEPR: 73.7%).

*1) Temporal Stereo Matching:* The static stereo method in [1] can hardly recover accurate depth of structures that are parallel to the baseline of the stereo camera. This is because the spatio-temporal profile of these structures is not distinctive along the baseline, and thus, multiple false-positive matches will be obtained during the block matching operation. Inspired by [76], we introduce the temporal stereo method (i.e., establishing stereo associations between adjacent views of the same camera) to resolve this problem. It is known that as long as the stereo camera does not move in the direction of the baseline, epipolar lines defined between a temporal-stereo pair are no longer parallel to the static-stereo baseline. This assumption always holds for forward moving stereo cameras, e.g., those employed in driving scenarios.

Let us consider the commonly used horizontal stereo configuration. We divide the sampled contour points from Section IV-A into two groups according to their gradient direction on the TS. In the first group, we collect pixels at which the ratio ($\eta$) between the vertical gradient and the horizontal gradient is smaller than

[2]By "TEPR," we mean the proportion of contour points covered by the sampled points, as illustrated in Fig. 5.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

a threshold. These pixels are fed to the static stereo method. The remaining sampled pixels go to the second group and are fed to the proposed temporal stereo method.

The key to the event-based temporal-stereo problem is to effectively exploit appearance similarity in event data. This requires the applied event representation, on which the stereo data association is established, to possess the speed-invariant property to some extent. Since the TS used in ESVO [1] does not satisfy this property, it is not a good choice for temporal stereo matching. To address this, we investigate the temporal stereo matching performance on three representations, namely, TOS [73], SILC [72], and our AA. We find that AA is the optimal choice due to its highest contour points ratio. Consequently, the epipolar matching is carried out on successive AA maps of the left camera, followed by triangulation and optionally by nonlinear refinement.

Note that the complementary property of the temporal-stereo configuration to the static one would hold as long as the temporal-stereo epipolar line is not parallel to the static-stereo baseline. This assumption is typically valid for most mobile robot platforms (e.g., ground vehicles and drones) when the stereo camera is set looking forward.

*2) Fast Block Matching:* The static stereo method in ESVO [1] is initialized by a block-matching operation, which uses the zero-mean normalized cross-correlation (ZNCC) as a similarity measure as follows:

$$Z_{x,y,d} = \frac{\text{cov}_{x,y,d}(l,r)}{\sqrt{\text{var}_{x,y}(l)\ \text{var}_{x,y+d}(r)}} \tag{4}$$

where $\text{cov}_{x,y,d}(l,r)$ denotes the intensity covariance between the patch $p_l$ centered at $(x,y)$ on image $l$ and the patch $p_{r,d}$ centered at $(x,y+d)$ on image $r$; $\text{var}_{x,y}(l)$ and $\text{var}_{x,y+d}(r)$ refer to the variance of $p_l$ and $p_{r,d}$, respectively.

Although the original ZNCC leads to satisfactory accuracy, it is notably time consuming. To speed up, we propose a fast version of ZNCC based on the approach presented in [77]. It adopts a recursive method that calculates the required sums and squared sums for the computation of ZNCC as follows:

$$\text{cov}_{x,y,d}(l,r) = \frac{\langle p_l, p_{r,d}\rangle - \mu_l T_{r,d}}{mn} \tag{5}$$

$$\text{var}_{x,y,d}(r) = \frac{mnT_{r^2,d} - T_{r,d}T_{r,d}}{(mn)^3} \tag{6}$$

$$T_{r,d} = T_{r,d-1} - \sum_{i=1}^{m} p_{r,d-1}(i,1) + \sum_{i=1}^{m} p_{r,d}(i,n) \tag{7}$$

$$T_{r^2,d} = T_{r^2,d-1} - \sum_{i=1}^{m} (p_{r,d-1}(i,1))^2 + \sum_{i=1}^{m} (p_{r,d}(i,n))^2 \tag{8}$$

where $m$ and $n$ denote the number of rows and columns of $p_l$ and $p_{r,d}$, $\mu_l$ the mean intensity of $p_l$, $T_{r,d}$ the sum of the intensities in $p_{r,d}$, and $T_{r^2,d}$ the sum of the squared intensities in $p_{r,d}$. In this way, the approach effectively utilizes intermediate results from previous computations to reduce redundant computation in ZNCC. As shown in Table III, this recursive method significantly reduces the computational complexity of block matching, leading to a speedup of approximately an order of magnitude.

TABLE III
COMPUTATIONAL COMPLEXITY OF DIFFERENT IMPLEMENTATIONS FOR ZNCC

| Algorithm | Traditional ZNCC | Fast ZNCC |
|---|---|---|
| Mean time | 30.70 $\mu$s | 5.761 $\mu$s |
| Median time | 30.16 $\mu$s | 5.714 $\mu$s |
| 25th percentile | 29.97 $\mu$s | 5.571 $\mu$s |
| 75th percentile | 30.45 $\mu$s | 5.819 $\mu$s |
| Standard deviation | 2.369 $\mu$s | 0.452 $\mu$s |

The test is conducted on four time surfaces selected from sequences of different resolutions, and 300 blocks are selected on each time surface. Block size is set to 15×15 px, with a search distance of 100 px.

*3) Depth Merge:* Given as input the relative pose between successive AA maps of the left camera, the proposed temporal-stereo method returns the depth information of structures lacking in the static-stereo method. Ideally, these stereo results can be fused in the same way as done in [1], because the temporal-stereo residuals $r_{\text{temporal}}$ evaluated on AAs and the static-stereo residuals $r_{\text{static}}$ evaluated on TSs approximately obey the *Student's t* distribution

$$r \sim St(\mu_r, s_r, \nu_r) \tag{9}$$

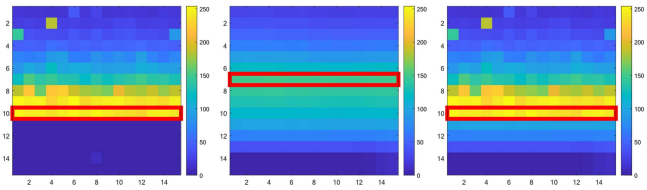
where $\mu_r$, $s_r$, and $\nu_r$ are the model parameters, namely, the mean, scale, and degree of freedom. However, this is not sensible because the uncertainty of the temporal-stereo results is always much smaller, leading to a bias toward the temporal-stereo estimates. Hence, we straightforwardly merge the results of the two stereo methods to obtain a more complete depth map, as shown in Section VI-B. Specifically, the employed merging operation consists of two steps. First, the depth estimates from the static-stereo and temporal-stereo methods are fused independently into two separate depth maps according to their respective probabilistic characteristics, in the same way as described in [1]. Afterward, we fill the nonoccupied pixels in the static-stereo depth map with the results in the temporal-stereo depth map. Also note that we get rid of the nonlinear refinement of depth in the static-stereo and temporal-stereo methods in our implementation. We find that the absence of the nonlinear refinement guarantees the real-time performance of mapping, while not bringing any notable degradation in the accuracy of camera pose tracking. The effect of omitting the nonlinear refinement is discussed in Section VI-E3.

### C. Localization: Camera Pose Estimation

Our front-end follows the tracking module in ESVO [1], which takes as input a TS and a local 3-D map, and computes the pose of the stereo rig with respect to the map. Let $\mathcal{S}^{\mathcal{F}_{\mathbf{ref}}} = \{\mathbf{x}_i\}$ represent a set of pixels with inverse depth values $\{\rho_i\}$ in the reference frame, and $\mathcal{T}(\mathbf{x},t)$ and $\overline{\mathcal{T}}(\mathbf{x},t) = 1 - \mathcal{T}(\mathbf{x},t)$ be the TS and negative TS at time $t$. The objective is, as in [1], to determine the optimal motion parameters $\boldsymbol{\theta}$ by solving

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \sum_{\mathbf{x}_i \in \mathcal{S}^{\mathcal{F}_{\mathbf{ref}}}} \overline{\mathcal{T}}_{left}(\boldsymbol{W}(\mathbf{x}_i, \rho_i; \boldsymbol{\theta})) \tag{10}$$

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 6. *Comparison of the original TS, Gaussian-blurred TS, and offset-free smoothed TS (OS-TS) for the same patch.* The color from blue to yellow represents an increase in pixel value. The pixel row with the highest values is highlighted using a red rectangle. Real data from *dsec_city_04a* are used in rendering these three representations. (a) Original TS. (b) Blurred TS. (c) Offset-free TS.

**Algorithm 2:** Offset-free Smoothed Time Surface (OS-TS).

**Input:** Original time surface $\mathrm{TS}_{ori}$.
**Output:** Offset-free smoothed time surface $\mathrm{TS}_{offset_free}$.

```
TS_blur ← Apply Gaussian blur to TS_ori.
for each pixel (i, j) in TS_ori do
  if TS_ori(i, j) == 0 then
    TS_offset_free(i, j) = TS_blur(i, j).
  else
    TS_offset_free(i, j) = TS_ori(i, j).
  end if
end for
```

where the warp $\boldsymbol{W}(\mathbf{x}_i, \rho_i; \boldsymbol{\theta}) = \pi_{\mathrm{left}}(T(\pi_{\mathrm{ref}}^{-1}(\mathbf{x}_i, \rho_i), G(\boldsymbol{\theta})))$ transfers points from the reference frame $\mathcal{F}_{\mathbf{ref}}$ to the latest TS using the inverse depth $\rho_i$ and projection operator $\pi$ (see [1]). The transformation matrix $G(\boldsymbol{\theta}) : \mathbb{R}^6 \mapsto SE(3)$ corresponds to the motion parameters $\boldsymbol{\theta} \doteq (\mathbf{c}^\top, \mathbf{t}^\top)^\top$, where $\mathbf{c} = (c_1, c_2, c_3)^\top$ are the Cayley parameters [78] for rotation and $\mathbf{t} = (t_x, t_y, t_z)^\top$ are translation parameters.

The spatio-temporal registration module leverages the distance-transform property of time surfaces and formulates the camera pose estimation as a 3-D-2-D registration problem. The employment of time surfaces smoothed with a blur kernel, on the one hand, enhances the differentiability of the objective function. On the other hand, the true location of edge points is shifted on the image plane, leading to a bias in the registration result. The nonlinear property of the resulting 3-D-2-D registration problem must be initialized properly, especially in the presence of a sudden variation in the orientation. Also, it may come across a degeneration in recovering the pitch and yaw components, and thus, an additional sensor (e.g., an IMU) is needed as a supplement. We discuss our solutions to these problems in the following.

*1) Offset-Free Smoothed Time Surface (OS-TS):* The original TS provides a truncated distance field with slopes ascending toward the true location of contours / edges. The lack of gradients in the truncated area, as shown in Fig. 6(a), leads to a one-sided convergence basin. To this end, a blurring filter is applied to the TS in ESVO [1] to obtain a two-sided convergence basin, as illustrated in Fig. 6(b). The smoothing operation, however, shifts the basin, leading to a notable offset in the location of the edges. This phenomenon can be seen by comparing the region with the highest pixel values in Fig. 6(a) and (b). Consequently, the spatio-temporal registration with blurred TS is no longer optimal because the 3-D map is aligned with the shifted observation of edges.

To address this issue, we propose an alternative design, called offset-free smoothed time surface (OS-TS), as outlined in Algorithm 2. Specifically, it fills the side lacking valid gradient information in the original TS [see Fig. 6(a)] using the pixel values in the corresponding area of the smoothed TS [see Fig. 6(b)]. As illustrated in Fig. 6(c), the result exhibits no offset while preserving valid gradient information on both sides of the edge. Finally, the negative OS-TS is computed and used in the spatio-temporal registration.

*2) Using IMU Preintegration as Motion Priors:* To deal with sudden variations in orientation and provide a proper initialized value for the spatio-temporal registration, we use as motion priors the preintegration of an IMU [79]. For an IMU with a 3-axis accelerometer and gyroscope, the measurements of acceleration $\tilde{\boldsymbol{a}}^b$ and angular velocity $\tilde{\boldsymbol{\omega}}^b$ can be denoted as follows:

$$\begin{aligned} \tilde{\boldsymbol{a}}^b &= \boldsymbol{a}^b + \mathbf{b}_a^b + \mathbf{n}_a^b \\ \tilde{\boldsymbol{\omega}}^b &= \boldsymbol{\omega}^b + \mathbf{b}_g^b + \mathbf{n}_g^b \end{aligned} \tag{11}$$

where $\mathbf{b}_a^b$, $\mathbf{b}_g^b$, $\mathbf{n}_a^b$, and $\mathbf{n}_g^b$ are the bias and noise of the accelerometer and gyroscope in the IMU frame, respectively. The IMU measurement, e.g., from time $t_i$ to $t_{i+1}$, expressed in the IMU frame $b_i$, can be calculated as follows:

$$\begin{aligned} \hat{\boldsymbol{\alpha}}_{b_{i+1}}^{b_i} &= \iint_{t\in[t_i,t_{i+1}]} \mathbf{R}_{b_t}^{b_i}\left(\boldsymbol{a}^{b_t}\right) dt^2 \\ \hat{\boldsymbol{\beta}}_{b_{i+1}}^{b_i} &= \int_{t\in[t_i,t_{i+1}]} \mathbf{R}_{b_t}^{b_i}\left(\boldsymbol{a}^{b_t}\right) dt \\ \hat{\boldsymbol{\gamma}}_{b_{i+1}}^{b_i} &= \int_{t\in[t_i,t_{i+1}]} \frac{1}{2}\boldsymbol{\Omega}\left(\boldsymbol{\omega}^{b_t}\right) \hat{\boldsymbol{\gamma}}_{b_t}^{b_i} dt \end{aligned} \tag{12}$$

where $\boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} -\lfloor\boldsymbol{\omega}\rfloor_\times & \boldsymbol{\omega} \\ -\boldsymbol{\omega}^\top & 0 \end{bmatrix}$, and $\lfloor\cdot\rfloor_\times$ is the cross-product matrix operator of a vector. The linear velocity and biases (i.e., $\mathbf{v}_{b_i}$, $\mathbf{b}_a^b$, and $\mathbf{b}_g^b$) are initialized empirically and constantly updated in the back-end (see Section V). We estimate the position prior from $t_i$ to $t_{i+1}$ by $(t_{i+1} - t_i)\mathbf{v}_{b_i}$, while neglecting the impact of acceleration due to the high operating rate of our tracking module, and employ $\hat{\boldsymbol{\gamma}}_{b_{i+1}}^{b_i}$ as rotation prior. The prior position and orientation are used to initialize the search for the optimal pose in (10).

## V. Back-End Optimization

To suppress the drift in the motion estimates, a bundle adjustment (BA) optimization is typically needed as a back-end. One of the most widely used strategies is the sliding-window approach, which jointly refines the poses of all involved key frames and, optionally, the depth information of covisible 3-D points by minimizing the inconsistency among heterogeneous measurements. However, our geometric method lacks explicit data association in the spatio-temporal domain, and thus, such a

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

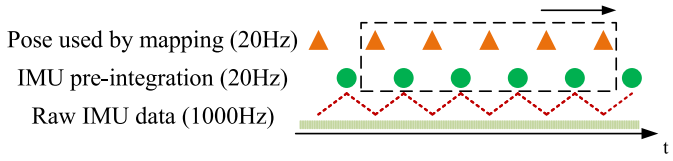



Fig. 7. *Illustration of data within the sliding window.* The data includes five poses used by mapping (orange triangles) and four results of the interframe IMU preintegration (green circles).

relaxation-and-refinement scheme [80] is not applicable. Moreover, we observe that the presence of camera poses among the optimization variables does not improve the camera tracking performance. This is because the local map (i.e., the involved point cloud within the window) is a fusion result relying on existing camera poses. It is by no means to establish additional constraints by introducing the 3-D point cloud into the sliding-window optimization without relaxation. Therefore, unlike [63] and [81], we attempt to create a compact and efficient back-end, which considers only the linear velocity and IMU's bias as the optimizing variables. Experiments show that the back-end built on this strategy significantly improves the tracking accuracy of our system (see Section VI-E).

Specifically, we apply a sliding window (shown in Fig. 7), in which the last $N$ pose estimates[3] and the corresponding IMU's measurements are involved. The objective is to refine the estimated linear velocity and IMU biases at the corresponding time of each involved pose by minimizing the Mahalanobis norm of IMU measurement residuals as follows:

$$\boldsymbol{\chi}^* = \arg\min_{\boldsymbol{\chi}} \sum_{i\in\mathcal{I}} \| r_{\mathcal{I}}\left(\hat{\boldsymbol{z}}^{b_i}_{b_{i+1}}, \boldsymbol{\chi}, \boldsymbol{\zeta}\right) \|^2_{\mathbf{P}^{b_i}_{b_{i+1}}} \tag{13}$$

where $\boldsymbol{\chi}^*$ denotes the optimizing variables (i.e., the linear velocities and the biases of IMU), $\boldsymbol{\zeta}$ the involved pose estimates, and $\hat{\boldsymbol{z}}$ the IMU preintegrated measurements. In addition, $\mathcal{I}$ denotes the set of IMU preintegration results in the sliding window. Furthermore, $\mathbf{P}^{b_i}_{b_{i+1}}$ is the covariance matrix of the IMU preintegration during the propagation process, and it is used as the weight for the IMU measurements. Detailed derivation can be found in [81, Sec. IV.B]. The relative pose residual $r_{\mathcal{I}}(\hat{\boldsymbol{z}}^{b_i}_{b_{i+1}}, \boldsymbol{\chi}, \boldsymbol{\zeta})$ is specifically defined as follows:

$$\begin{aligned} r_{\mathcal{I}}(\hat{\boldsymbol{z}}^{b_i}_{b_{i+1}}, \boldsymbol{\chi}, \boldsymbol{\zeta}) &= \left[\delta\boldsymbol{\alpha}^{b_i\top}_{b_{i+1}}, \delta\boldsymbol{\beta}^{b_i\top}_{b_{i+1}}, \delta\boldsymbol{\gamma}^{b_i\top}_{b_{i+1}}, \delta\mathbf{b}^\top_a, \delta\mathbf{b}^\top_g\right]^\top \\ &= \begin{bmatrix} \mathbf{R}^{b_i}_w \left(\mathbf{p}^w_{b_{i+1}} - \mathbf{p}^w_{b_i} + \frac{1}{2}\mathbf{g}^w\Delta t_i^2 - \mathbf{v}^w_{b_i}\Delta t_i\right) - \hat{\boldsymbol{\alpha}}^{b_i}_{b_{i+1}} \\ \mathbf{R}^{b_i}_w \left(\mathbf{v}^w_{b_{i+1}} + \mathbf{g}^w\Delta t_i - \mathbf{v}^w_{b_i}\right) - \hat{\boldsymbol{\beta}}^{b_i}_{b_{i+1}} \\ 2\left[(\mathbf{q}^w_{b_i})^{-1} \otimes \mathbf{q}^w_{b_{i+1}} \otimes \left(\hat{\boldsymbol{\gamma}}^{b_i}_{b_{i+1}}\right)^{-1}\right]_{xyz} \\ \mathbf{b}^{b_{i+1}}_a - \mathbf{b}^{b_i}_a \\ \mathbf{b}^{b_{i+1}}_g - \mathbf{b}^{b_i}_g \end{bmatrix} \end{aligned} \tag{14}$$

where $\mathbf{g}^w$ indicates the gravity in the world coordinate system, $\mathbf{p}$ and $\mathbf{q}$ denote the position and rotation, respectively. $[\cdot]_{xyz}$ extracts the vector portion of the quaternion $\mathbf{q}$ for representing the error state. In detail, the size of the sliding window is set to 5, and we apply the Levenberg–Marquardt (LM) algorithm to solve the optimization problem. The sliding window shifts forward when a new pose estimate is obtained, while the oldest pose estimate and corresponding IMU measurements are removed.

[3]Note that the involved poses are only used for providing the relative pose constraints rather than being updated.

TABLE IV
EVENT CAMERA PARAMETERS OF THE DATASETS USED IN THE EXPERIMENTS

| **Dataset** | **Cameras** | **Resolution** [px] | **BL** [cm] | **FOV** [°] |
|---|---|---|---|---|
| *rpg* [85] | DAVIS240C | 240×180 | 14.7 | 62.9 |
| *MVSEC* [84] | DAVIS346 | 346×260 | 10.0 | 74.8 |
| *DSEC* [82] | Prophesee Gen3.1 | 640×480 | 59.9 | 60.1 |
| *VECtor* [83] | Prophesee Gen3 | 640×480 | 17.0 | 67.0 |
| *TUM-VIE* [86] | Prophesee Gen4 | 1280×720 | 11.8 | 65.0 |
| *Ours* (Sec. VI-D) | DVXplorer | 640×480 | 51.0 | 72.4 |

## VI. EXPERIMENTS

In this section, we comprehensively evaluate our system. First, we introduce all datasets used and explain why some of them are not applicable for the evaluation (see Section VI-A). Second, we conduct both qualitative and quantitative comparisons of the proposed mapping module against that of ESVO, justifying the benefits brought by our modifications (see Section VI-B). Third, we evaluate the overall performance of our system by comparing the localization results against those of various stereo event-based pipelines and demonstrate our superior performance in terms of trajectory accuracy, especially in driving scenarios (see Section VI-C). Besides, an extended evaluation using our dataset is provided to justify the versatility of our system (see Section VI-D). Moreover, we investigate the benefits brought by the newly incorporated functions and modules on the tracking performance (see Section VI-E) and provide a computational-complexity analysis of each function in the entire system (see Section VI-F). Finally, we discuss some limitations of our method in small-scale and narrow indoor scenes (see Section VI-G).

### A. Public Datasets Used

Our system is tested on five publicly available datasets with stereo event cameras and IMUs, which feature different event-camera types, spatial resolutions, and scene geometries, including *DSEC* [82], *VECtor* [83], *MVSEC* [84], *rpg* [85], and *TUM-VIE* [86]. *DSEC* is a large-scale dataset capturing outdoor driving scenes, and the others feature small indoor scenes with different sensor resolutions. Sequences in *MVSEC* are recorded using a stereo event camera mounted on drones, whereas a hand-held rig is used in *VECtor*, *rpg*, and *TUM-VIE*. The characteristics of each dataset are summarized in Table IV.

Note that some of the other released datasets using stereo event cameras are not applicable to our evaluation. For example, an unsatisfactory rectification is witnessed in the *HKU* dataset [2] due to inaccuracies in the extrinsic calibration. The *ECMD* dataset [87] exhibits a significant discrepancy in the streaming rate of events between the left and right event cameras, thus

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

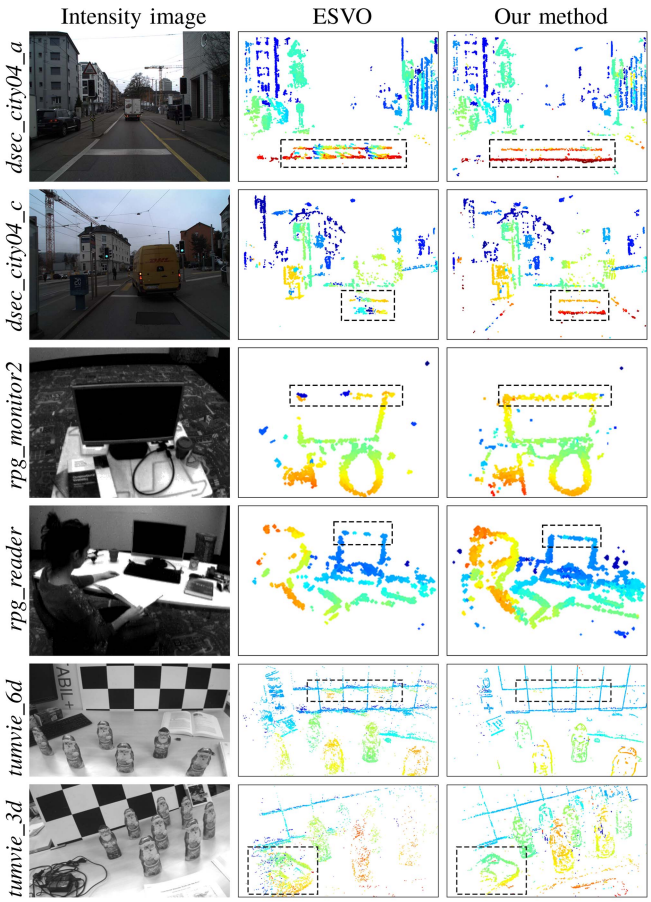



Fig. 8. *Qualitative comparison of mapping results*. Intensity images in the first column are used only for visualization. The second and third columns show the estimated inverse depth maps by ESVO [1] and our method, respectively. Note that our method returns more accurate and complete reconstruction results for horizontal edges. Inverse depth maps are color coded, from red (close) to blue (far) over a white background, in the range of 0.5 m–10 m for the *rpg* dataset, 1 m–50 m for the *DSEC* dataset, and 0.33 m–10 m for the *TUM-VIE* dataset.

breaking the assumption of temporal coherence between the two cameras. Last but not the least, the "Kirschbaum problem" [88] is witnessed in the *M3ED* dataset [89], which employs event cameras with a spatial resolution of 1280 × 720 pixels and collects data in urban environments filled with high-frequency texture patterns (e.g., trees and lawns). According to [90], the higher the spatial resolution, the more sensitive each pixel becomes to brightness changes. Therefore, such a sensor-scene configuration results in an excessive number of events recorded in this dataset, leading to a lack of clear structure information in the spatio-temporal profile of events. This may result in failures of the proposed direct method.

### B. Comparison of Mapping: ESVO versus Ours

We compare our mapping results against those of the original ESVO pipeline [1] to demonstrate the performance improvement of our system in mapping. As illustrated in Fig. 8, the results of ESVO exhibit that horizontal structures in the environment are either not reconstructed or inaccurate in depth estimates. This is because the static-stereo method cannot precisely recover the depth of structures that are parallel to the stereo-camera baseline. In comparison, our system additionally incorporates a temporal stereo method, which allows for accurate depth estimation of structures parallel to the stereo-camera baseline (see Section IV-B1). Besides, the sampled points from AA better capture edge structures with less noise (see last row of Fig. 8). All of these lead to better performance in terms of reconstruction completeness and local smoothness.

TABLE V
QUANTITATIVE COMPARISON OF MAPPING RESULTS

| Sequence | Depth range | Statistic | ESVO | Ours |
|---|---|---|---|---|
| *dsec_city04_a* | 11.54 m | Mean error | 1.84 m | **1.64 m** |
| | | Median error | 1.45 m | **1.29 m** |
| | | Relative error | 15.15% | **12.57%** |
| *dsec_city04_c* | 9.75 m | Mean error | 1.22 m | **0.99 m** |
| | | Median error | **0.37 m** | 0.42 m |
| | | Relative error | 11.45% | **9.53%** |
| *dsec_city04_d* | 11.37 m | Mean error | 1.72 m | **1.59 m** |
| | | Median error | 1.83 m | **1.35 m** |
| | | Relative error | 13.52% | **11.20%** |

Depth range refers to the mean of true depth of the evaluated points.

We also provide a quantitative evaluation, where the figures of merit consist of the *mean* depth error (defined as $\frac{1}{N}\sum_i^N |d_{i,\text{est}} - d_{i,\text{gt}}|$), the *median* depth error, and the relative error (defined as $\frac{1}{N}\sum_i^N |d_{i,\text{est}} - d_{i,\text{gt}}|/d_{i,\text{gt}}$), respectively. Sequences that frequently capture horizontal structures are selected for the evaluation. Depth measurements from a LiDAR are adopted as ground truth, and only those within 20 m are used in the evaluation because we find that some of them beyond that range are either less accurate or not aligned well with visual data. As shown in Table V, the best results under each metric are highlighted in bold, demonstrating that our method outperforms ESVO in almost all evaluation metrics.

### C. Full System Evaluation

To assess the performance of camera pose tracking of our system, we conduct tests using all five datasets of different resolutions and compare the motion estimation results with another five stereo event-based VO/VIO pipelines [1], [2], [42], [63], [66]. Among them, ESVO [1] is the first stereo event-based VO system that employs a direct method. Successive works, e.g., ESVIO [63] and our ICRA publication [42], are both direct methods that additionally introduce an IMU. Note that a feature-based method, presented in [2], leverages both standard cameras and event cameras. To ensure a fair comparison, only the pure event-based part (denoted by ESIO [2]) is enabled and used as a comparative. Event-based stereo parallel tracking and mapping (ES-PTAM) [66] is the latest event-only direct stereo VO system, with its mapping module built upon multicamera event-based multiview stereo (MC-EMVS) [67]. Note that some systems only provide raw data over several sequences. Therefore, we can only use the provided trajectories in our comparison and denote the missing ones with "-" in the tables. We use "failed" to denote cases where the open-sourced systems failed to run successfully.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABSOLUTE ROTATION ERROR AND ATE (RMS) ON DATASETS OF MULTIPLE RESOLUTIONS [ARE:°, ATE:CM]

| Dataset | Sequence | ESVO [1] | | ES-PTAM [66] | | ESIO [2] | | ESVIO [63] | | ICRA'24 [42] | | Ours | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ARE | ATE | ARE | ATE | ARE | ATE | ARE | ATE | ARE | ATE | ARE | ATE |
| *rpg* | *box* | 5.95 | 5.80 | 6.62 | **4.06** | 12.35 | 11.38 | 4.25 | 4.41 | 3.30 | 6.67 | **2.79** | 4.31 |
| | *monitor* | 6.21 | 3.30 | **1.52** | 2.34 | 17.67 | 7.87 | 3.79 | 3.48 | 3.53 | 2.80 | 2.19 | **2.31** |
| | *bin* | 1.98 | 2.80 | 3.29 | 2.57 | 11.76 | 7.08 | 3.87 | 2.28 | 3.54 | 5.90 | **1.23** | **2.27** |
| | *desk* | 10.49 | 3.20 | **3.44** | 2.84 | 3.60 | 3.16 | 7.09 | 2.03 | 6.72 | 5.33 | 4.11 | **1.57** |
| | *reader* | 3.80 | 6.60 | – | – | failed | failed | – | – | 1.78 | 3.88 | **1.54** | **2.68** |
| *MVSEC* | *indoor_1* | 4.40 | 16.59 | 14.93 | 15.02 | 22.32 | 820.36 | 6.57 | 9.63 | 11.16 | 17.65 | **1.69** | **7.63** |
| | *indoor_2* | 5.69 | 14.94 | – | – | 43.49 | 417.85 | – | – | 12.14 | 17.52 | **4.53** | **10.05** |
| | *indoor_3* | 2.94 | 10.03 | – | – | failed | failed | 3.01 | 8.06 | 2.73 | 10.45 | **2.63** | **7.35** |
| | *indoor_4* | failed | failed | – | – | 32.15 | 173.51 | – | – | failed | failed | **10.96** | **5.59** |
| *DSEC* | *city04_a* | 8.17 | 370.32 | **3.17** | 131.62 | 5.07 | 940.80 | 5.59 | 201.53 | 4.36 | 103.85 | 3.33 | **56.17** |
| | *city04_b* | 1.97 | 115.56 | 2.04 | 29.02 | 3.69 | 434.87 | **1.38** | **48.33** | 2.64 | 66.80 | 1.55 | 73.83 |
| | *city04_c* | 14.04 | 932.84 | 6.02 | 1184.37 | **7.35** | 1153.69 | 14.62 | 1400.76 | 11.39 | 637.13 | 10.26 | **508.71** |
| | *city04_d* | 21.62 | 2676.11 | 37.13 | 1053.87 | **3.63** | 6822.53 | – | – | 14.14 | 732.13 | 8.87 | **546.58** |
| | *city04_e* | 6.89 | 792.93 | 3.97 | 75.90 | 5.57 | 1036.23 | 6.28 | 331.70 | 5.08 | 115.82 | **3.54** | **52.94** |
| | *city04_f* | 6.36 | 1400.26 | 10.65 | 522.00 | 6.19 | 4595.01 | 20.78 | 1765.48 | 5.52 | 579.66 | **5.00** | **257.75** |
| | *city09_b* | 5.17 | 606.09 | **1.75** | 195.14 | failed | failed | – | – | 2.92 | 192.44 | 2.17 | **87.83** |
| | *city11_a* | 1.96 | 366.22 | – | – | 2.40 | 107.36 | 7.71 | 406.11 | 4.52 | 95.75 | **1.85** | **48.77** |
| | *city11_b* | 44.57 | 3241.69 | – | – | **3.13** | **300.14** | – | – | 12.78 | 869.77 | 10.61 | 441.79 |
| *VECtor* | *robot_normal* | 19.79 | 7.32 | – | – | 5.61 | 5.17 | – | – | 20.46 | 15.20 | **5.03** | **4.81** |
| | *robot_fast* | failed | failed | – | – | failed | failed | – | – | failed | failed | **16.29** | **24.18** |
| | *corner_slow* | 9.63 | 13.70 | – | – | 32.67 | 2.67 | – | – | 6.15 | 5.52 | **2.96** | **2.15** |
| | *hdr_normal* | 24.54 | 18.40 | – | – | 18.42 | 27.85 | – | – | 14.11 | 16.06 | **7.82** | **13.53** |
| | *sofa_normal* | failed | failed | – | – | **12.11** | 43.94 | – | – | failed | failed | 20.11 | **40.28** |
| | *desk_normal* | 19.91 | 20.81 | – | – | failed | failed | – | – | 10.36 | 19.08 | **6.82** | **16.47** |
| *TUM-VIE* | *1d_trans* | 13.47 | 12.54 | **6.02** | **1.05** | failed | failed | – | – | 6.67 | 3.86 | 6.30 | 3.33 |
| | *3d_trans* | 19.20 | 17.19 | 15.62 | 8.53 | failed | failed | – | – | 17.93 | 18.90 | **6.61** | **7.26** |
| | *6d_trans* | 17.59 | 13.46 | 14.01 | 10.25 | failed | failed | – | – | failed | failed | **4.17** | **3.21** |
| | *desk* | 14.56 | 12.92 | 3.37 | **2.50** | failed | failed | – | – | 6.95 | 8.99 | **2.40** | 6.16 |
| | *desk2* | 5.86 | 4.42 | 10.12 | 7.20 | failed | failed | – | – | 4.32 | 9.47 | **3.88** | **4.02** |

ESVO and ES-PTAM are event-only methods, whereas the rest process input stereo events and inertial (IMU) data.
* *"-" represents the lack of experimental results of the algorithm (according to the respective publication).*

To quantitatively evaluate the motion estimation results, we use two standard evaluation metrics: relative pose error (RPE) and absolute trajectory error (ATE) [92]. The results of all methods can be found in Tables VI and VII, in which the best results are highlighted in bold. In general, the proposed method ("Ours") outperforms the others in terms of RPE and ATE. Our method resolves most of the remaining issues of ESVO [1], thus leading to more accurate trajectory estimates than ESVO.

In contrast to ESVIO [63], our method advances mainly in the preprocessing and mapping module. The proposed contour-point sampling method and the additionally introduced temporal-stereo operation significantly improve the mapping performance, which benefits the entire system. In addition, our better results justify that having the spatio-temporal-registration residual term incorporated into the back-end by ESVIO [63] does not bring obvious benefits to motion estimation while causing additional computational burden. This may explain why ESVIO [63] does not demonstrate a real-time performance on event data with VGA resolution.

Compared with ESIO [2], the only feature-based method in the table, our direct method makes better use of event data, circumventing the theoretical flaw of event-based feature matching. This advantage of direct methods over feature-based methods is obvious in the results for sequences collected in large-scale environments (e.g., *DSEC* in Table VI). Notably, ESIO cannot operate properly on the *TUM-VIE* dataset. We believe that one possible reason is that the smoothing effect on the gradient profile of the event representation, caused by the high spatial resolution of the camera, significantly hinders the detection and tracking of ARC* [47], leading to the malfunction of the entire system.

In comparison to [42], our better performance is mainly attributed to the proposed back-end. The better estimated IMU biases and linear velocities at recent poses facilitate camera pose tracking, thus leading to a notable improvement in tracking accuracy. An in-depth discussion on the benefit brought by the back-end is given in Section VI-E1. Finally, ES-PTAM [66] performs exceptionally well in small-scale environments and those with repetitive textures, attributing it to the precise depth results via its multiview stereo mapping method. However, thanks to the effective utilization of IMU data in the back-end, our system significantly outperforms ES-PTAM in large-scale scenarios while being on par in small-scale environments.

In addition, ESVO2 demonstrates excellent performance in both high-speed motion and HDR scenes. The *vector_robot_fast* sequence is captured indoors with a stereo event camera under aggressive motion, and only our method can operate on it and achieve good results. Both *dsec_city_09b* and *vector_hdr_normal* are HDR sequences of nighttime driving and small-scale indoor scenarios, respectively. Benefiting from the accurate contour points sampling strategy, ESVO2 achieves the best performance on both of these sequences.

Figs. 9 and 10 provide a qualitative comparison of the trajectories estimated by ESVO and our method on sequences from *rpg*, *MVSEC*, and *DSEC* data (To deliver a compact and clear visualization, only these trajectories are displayed. A more

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

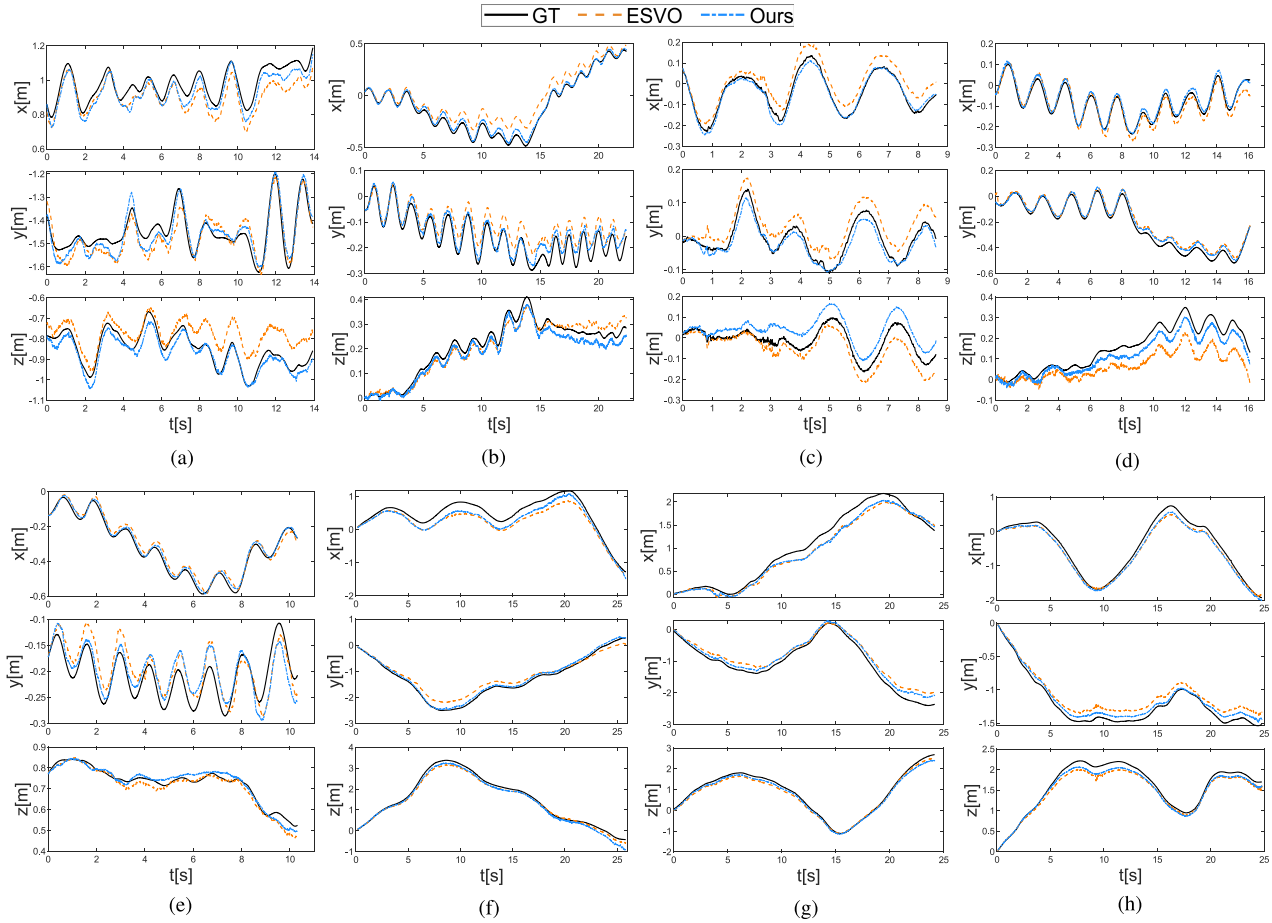



Fig. 9. Translation results on the *rpg* and *MVSEC* datasets, with ground truth provided by a motion capture system. (a) *rpg_box*. (b) *rpg_monitor*. (c) *rpg_reader*. (d) *rpg_bin*. (e) *rpg_desk*. (f) *upenn_indoor_flying1*. (g) *upenn_indoor_flying2*. (h) *upenn_indoor_flying3*.

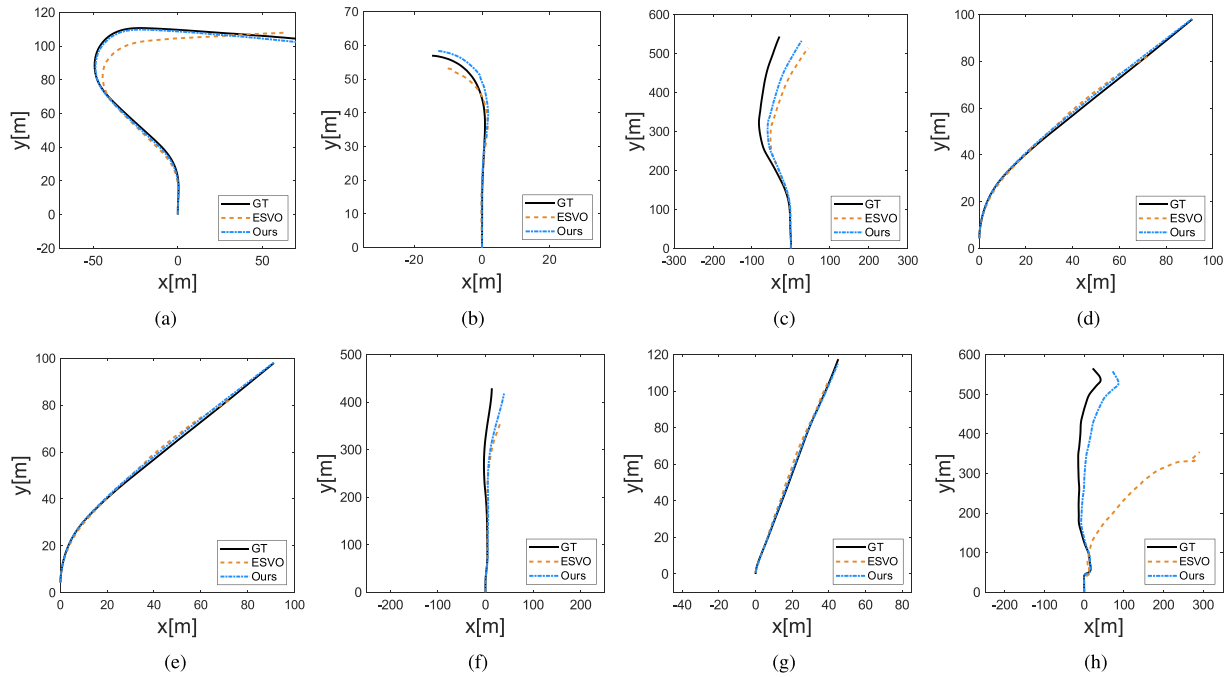



Fig. 10. Recovered trajectories on the *DSEC* dataset, with ground truth provided by a LiDAR-IMU-based method [91]. (a) *dsec_city04_a*. (b) *dsec_city04_b*. (c) *dsec_city04_c*. (d) *dsec_city04_d*. (e) *dsec_city04_e*. (f) *dsec_city04_f*. (g) *dsec_city11_a*. (h) *dsec_city11_b*.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
RPE (RMS) ON DATASETS OF MULTIPLE RESOLUTIONS [$\mathbf{R}$:°/s, $\mathbf{t}$:cm/s]

| Dataset | Sequence | ESVO [1] | | ES-PTAM [66] | | ESIO [2] | | ESVIO [63] | | ICRA'24 [42] | | Ours | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | **R** | **t** | **R** | **t** | **R** | **t** | **R** | **t** | **R** | **t** | **R** | **t** |
| *rpg* | *box* | 3.40 | 7.20 | **2.10** | 4.22 | 2.91 | 7.87 | 5.11 | 8.71 | 2.65 | 5.16 | 3.02 | **4.18** |
| | *monitor* | 1.70 | 3.20 | **1.14** | 2.59 | 1.71 | 4.27 | 6.12 | 7.93 | 1.29 | 1.85 | 1.15 | **1.69** |
| | *bin* | 1.20 | 3.10 | 1.00 | **1.82** | 2.30 | 6.78 | 4.14 | 7.64 | 1.23 | 3.80 | **0.94** | 2.53 |
| | *desk* | 3.10 | 4.50 | 1.77 | 3.99 | 3.94 | 3.79 | 8.56 | 3.65 | 2.60 | 6.33 | **1.76** | **3.53** |
| | *reader* | 2.50 | 5.60 | – | – | failed | failed | – | – | 1.37 | 4.65 | **1.32** | **2.12** |
| *MVSEC* | *indoor_1* | 1.09 | 7.38 | 1.68 | 6.89 | 2.91 | 228.84 | 1.28 | 5.92 | 2.09 | 10.72 | **0.98** | **5.05** |
| | *indoor_2* | 1.72 | 7.39 | – | – | 3.58 | 134.48 | – | – | 3.77 | 12.33 | **1.42** | **6.12** |
| | *indoor_3* | 1.08 | 5.97 | – | – | failed | failed | 0.92 | 4.81 | 0.99 | 6.05 | **0.77** | **4.75** |
| | *indoor_4* | failed | failed | – | – | 15.85 | 181.03 | – | – | failed | failed | **3.84** | **10.36** |
| *DSEC* | *city04_a* | 1.02 | 69.14 | **0.53** | 26.92 | 0.66 | 187.03 | 0.95 | 62.85 | 0.82 | 20.91 | 0.60 | **16.98** |
| | *city04_b* | 0.50 | 32.84 | 0.44 | 26.48 | 3.07 | 188.23 | 0.87 | 43.07 | 0.67 | 25.31 | **0.43** | **21.70** |
| | *city04_c* | 0.65 | 82.03 | 1.28 | 86.21 | 0.60 | 145.64 | 0.56 | 105.87 | 0.93 | 71.76 | **0.55** | **39.12** |
| | *city04_d* | 1.22 | 190.67 | 1.01 | 68.02 | 1.17 | 540.98 | – | – | 0.73 | 60.18 | **0.41** | **18.00** |
| | *city04_e* | 1.40 | 183.01 | **0.25** | **14.05** | 1.31 | 461.23 | 0.81 | 101.28 | 0.71 | 50.78 | 0.38 | 15.41 |
| | *city04_f* | 1.07 | 267.62 | 1.27 | 49.39 | 1.29 | 626.87 | 0.96 | 133.11 | 2.26 | 290.56 | **0.38** | **20.05** |
| | *city09_b* | 0.27 | 38.52 | **0.17** | 22.44 | failed | failed | – | – | 0.24 | 20.25 | 0.21 | **17.84** |
| | *city11_a* | 0.56 | 56.82 | – | – | 0.61 | 54.53 | 0.49 | 54.02 | 0.53 | 20.34 | **0.30** | **12.94** |
| | *city11_b* | 3.34 | 188.97 | – | – | 0.57 | 53.55 | – | – | 0.95 | 91.94 | **0.45** | **14.83** |
| *VECtor* | *robot_normal* | 6.72 | 16.79 | – | – | **2.08** | 3.38 | – | – | 4.92 | 6.18 | 2.36 | **3.16** |
| | *robot_fast* | failed | failed | – | – | failed | failed | – | – | failed | failed | **5.05** | **8.22** |
| | *corner_slow* | 4.07 | 6.45 | – | – | 1.93 | 7.01 | – | – | 1.71 | 3.16 | **1.34** | **1.91** |
| | *hdr_normal* | 2.27 | 5.55 | – | – | 3.96 | 15.65 | – | – | 2.44 | 6.54 | **1.53** | **3.79** |
| | *sofa_normal* | failed | failed | – | – | **2.03** | 54.32 | – | – | failed | failed | 7.15 | **13.59** |
| | *desk_normal* | 2.43 | 5.12 | – | – | failed | failed | – | – | 2.82 | 5.76 | **1.68** | **3.50** |
| *TUM-VIE* | *1d_trans* | 3.69 | 8.41 | **0.23** | **0.71** | failed | failed | – | – | 0.97 | 1.21 | 0.81 | 1.06 |
| | *3d_trans* | 3.41 | 7.47 | 2.58 | 4.94 | failed | failed | – | – | 1.79 | 3.98 | **0.83** | **2.32** |
| | *6d_trans* | 9.49 | 14.28 | 3.61 | 12.02 | failed | failed | – | – | failed | failed | **1.16** | **2.49** |
| | *desk* | 2.93 | 5.87 | 0.99 | **2.53** | failed | failed | – | – | 1.00 | 3.14 | **0.98** | 3.41 |
| | *desk2* | 2.12 | 6.23 | 2.98 | 4.04 | failed | failed | – | – | 2.15 | 4.20 | **0.91** | **2.41** |

ESVO and ES-PTAM are event-only methods, whereas the other methods process input stereo events and inertial (IMU) data.

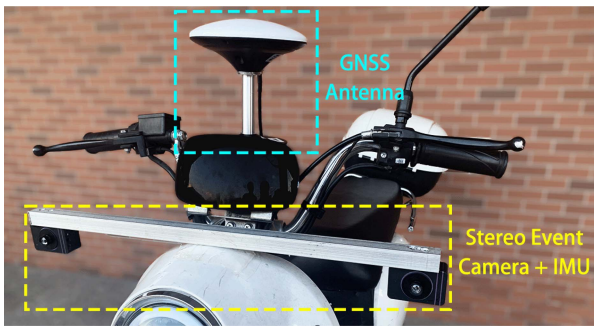



Fig. 11. Illustration of the applied sensor suite and the motorbike used for data collection.

complete qualitative comparison is provided in the Appendix). Since *DSEC* does not provide ground-truth trajectories, we use the result of a LiDAR-IMU-based method [91] as ground truth. In short, the trajectories obtained by our method align better with the ground truth, as evidenced by the quantitative analysis.

### D. Outdoor Evaluation

To further assess the versatility of our system, we collect our own outdoor dataset for evaluation. The employed sensor suite, shown in Fig. 11, consists of a stereo event camera (DVXplorer, 640 × 480 pixels) with a baseline distance of 51 cm, an embedded 6-axis IMU, and a u-blox ZED-F9P GNSS receiver. The GNSS receiver owns an internal RTK positioning engine that provides ground-truth trajectories for evaluation. All the sensors are calibrated and synchronized in advance.

Our outdoor dataset consists of two subsets (called *hnu_mapping* and *hnu_tracking*), which are used for assessing the performance of mapping and localization, respectively. The sequences in the *hnu_mapping* subset capture scenes that vary in depth range, texture richness, and brightness condition (including HDR scenarios), etc. Some of the mapping results can be found in Fig. 13, which demonstrate good mapping performance as has been witnessed on other publicly available datasets. The *hnu_tracking* subset consists of two sequences featuring a closed loop (*hnu_campus*, shown in Fig. 1) and a winding trajectory along a narrow street (*hnu_peachlake*, shown in Fig. 12), respectively. Compared to the other two stereo event-based visual odometry systems (i.e., ESVO and ESIO), the estimated trajectories of our system align better with the ground truth, especially exhibiting smaller drift in absolute orientation. More importantly, our system is the only one that can operate on these sequences in real time, demonstrating the potential for practical applications.

### E. Ablation Studies

We conduct several ablation studies to assess the benefits brought by the new functions and modules proposed in this work (see Sections VI-E1 to VI-E3). In addition, we investigate the sensitivity with respect to the size parameter in the sliding window and with respect to running our pipeline with/without the IMU input (see Section VI-E4). In all studies, we use the

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
RESULTS OF THE ABLATION STUDIES (SECTIONS VI-E1 TO VI-E3)

| Sequence | w/o back-end | | w/o OS-TS | | w/ refinement | | | Reference | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | ARE | ATE | ARE | ATE | ARE | ATE | Time | ARE | ATE | Time |
| *dsec_city04_a* | 3.86 ↑ | 1.34 ↑ | 4.77 ↑ | 1.58 ↑ | 3.01 ↓ | 0.59 ↑ | 44.70 ↑ (✗) | 3.33 | 0.56 | 32.50 (✓) |
| *dsec_city04_c* | 14.98 ↑ | 12.90 ↑ | 13.17 ↑ | 7.15 ↑ | 9.50 ↓ | 4.49 ↓ | 47.55 ↑ (✗) | 10.26 | 5.09 | 34.12 (✓) |
| *dsec_city04_d* | 9.83 ↑ | 5.89 ↑ | 14.87 ↑ | 8.20 ↑ | 7.67 ↓ | 5.37 ↑ | 47.03 ↑ (✗) | 8.87 | 5.04 | 33.45 (✓) |
| *dsec_city04_e* | 4.70 ↑ | 2.63 ↑ | 4.34 ↑ | 0.99 ↑ | 3.10 ↓ | 0.61 ↓ | 50.85 ↑ (✗) | 3.54 | 0.76 | 30.09 (✓) |
| *dsec_city04_f* | 5.31 ↑ | 7.81 ↑ | 5.40 ↑ | 3.45 ↑ | 4.70 ↓ | 3.21 ↓ | 46.41 ↑ (✗) | 5.00 | 3.28 | 32.14 (✓) |
| *dsec_city11_a* | 2.64 ↑ | 1.34 ↑ | 2.44 ↑ | 0.54 ↑ | 2.15 ↑ | 0.54 ↑ | 49.93 ↑ (✗) | 1.85 | 0.48 | 31.96 (✓) |
| *dsec_city11_b* | 16.31 ↑ | 9.02 ↑ | 13.94 ↑ | 8.01 ↑ | 9.42 ↓ | 4.60 ↑ | 47.86 ↑ (✗) | 10.61 | 4.41 | 36.77 (✓) |

The up/down arrows indicate the variation trend compared to the reference group. The "time" column demonstrates the average runtime of the mapping thread, followed with a tick or a cross indicating whether the mapping module can stably run in real time at a frequency of 20Hz. [ARE: °, ATE: M, Time: Ms].

TABLE IX
SENSITIVITY WITH RESPECT TO THE SIZE OF THE SLIDING WINDOW AND NECESSITY OF AN IMU (SECTION VI-E4)

| Sequence | Window size 8 | | Window size 12 | | w/o IMU | | Reference (size 5) | |
|---|---|---|---|---|---|---|---|---|
| | ARE | ATE | ARE | ATE | ARE | ATE | ARE | ATE |
| *dsec_city04_a* | 3.45 ↑ | 0.65 ↑ | 2.89 ↑ | 0.60 ↑ | 5.73 ↑ | 3.20 ↑ | 3.33 | 0.56 |
| *dsec_city04_c* | 12.23 ↑ | 5.49 ↑ | 11.71 ↑ | 4.77 ↓ | 18.02 ↑ | 14.78 ↑ | 10.26 | 5.09 |
| *dsec_city04_d* | 10.81 ↑ | 6.20 ↑ | 9.52 ↑ | 5.94 ↑ | 16.38 ↑ | 18.96 ↑ | 8.87 | 5.04 |
| *dsec_city04_e* | 3.45 ↓ | 0.63 ↓ | 3.36 ↓ | 0.60 ↓ | 5.06 ↑ | 2.95 ↑ | 3.54 | 0.76 |
| *dsec_city04_f* | 5.71 ↑ | 3.40 ↑ | 5.43 ↑ | 3.20 ↓ | 7.09 ↑ | 11.28 ↑ | 5.00 | 3.28 |
| *dsec_city11_a* | 2.47 ↑ | 0.66 ↑ | 2.42 ↑ | 0.61 ↑ | 3.85 ↑ | 1.11 ↑ | 1.85 | 0.48 |
| *dsec_city11_b* | 9.65 ↓ | 5.52 ↑ | 9.87 ↓ | 5.34 ↑ | 23.77 ↑ | 12.31 ↑ | 10.61 | 4.41 |

The up/down arrows indicate the variation trend compared to the reference group. [ARE: °, ATE: M].




Fig. 12. *Trajectory results on the hnu_peachlake sequence.* The ground-truth trajectory is provided by RTK (red) and aligned with Google Maps. Our method achieves the best performance in terms of global trajectory consistency compared to ESVO [1] (green) and ESIO [2] (yellow).

*DSEC* dataset [82] and compare the performance of our system (denoted as "reference" in Tables VIII and IX) with the following features disabled, respectively.

*1) Effect of the Back-End:* As said in Section V, the presence of back-end facilitates suppressing drift in IMU biases and predicting the linear velocity, thus leading to better performance in camera pose tracking. To justify this, we compare the trajectory results obtained with and without the back-end. In the configuration without the back-end (denoted by "w/o back-end" in Table VIII), the gyroscope preintegration results are still used as an initialization for the spatio-temporal registration, but its bias is not updated. As shown in Table VIII, "reference" exhibits notably fewer drifts in the estimated trajectories than "w/o back-end," demonstrating the necessity of constantly updating the belief of the IMU biases and the camera's linear velocity. With these updated parameters, we obtain better initial values for the spatio-temporal registration and, thus, more accurate motion estimation. Consequently, the improved tracking results lead to more accurate mapping results (especially for the temporal-stereo part) and, in turn, facilitate the tracking module.

It is also worth explaining the difference between the results in Table VI (denoted by "ICRA'24") and those in Table VIII. Note that the results obtained with the back-end disabled are less accurate than those reported in [42]. This is because we not only use the gyroscope's preintegration results in [42] but also predict the next position using a constant-velocity motion model. In this ablation study, only the rotation preintegration is performed, which indicates that the constant-velocity motion model can enhance the accuracy of trajectory estimation.

*2) Effect of OS-TS:* As claimed in Section IV-C, the proposed OS-TS ensures no offset in the location of edges while preserving valid gradient information in the neighboring area. To justify this, we compare the trajectory results obtained with and without OS-TS. In the configuration without OS-TS, the blurred TS [see Fig. 6(b)] is used in the 3-D-2-D spatio-temporal registration. As shown in Table VIII, the trajectory errors increase significantly when OS-TS is not employed (denoted by "w/o OS-TS"). This is because the highest pixel values in the blurred TS do not accurately correspond to the positions of the edges in the current scenes. As a result, even when the registration problem is solved successfully, the obtained pose estimation is consistently suboptimal. Our proposed OS-TS addresses this issue by filling in gradients on the original TS, ensuring the successful resolution of the registration problem and reducing the accumulated error caused by inaccurate pose estimations.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

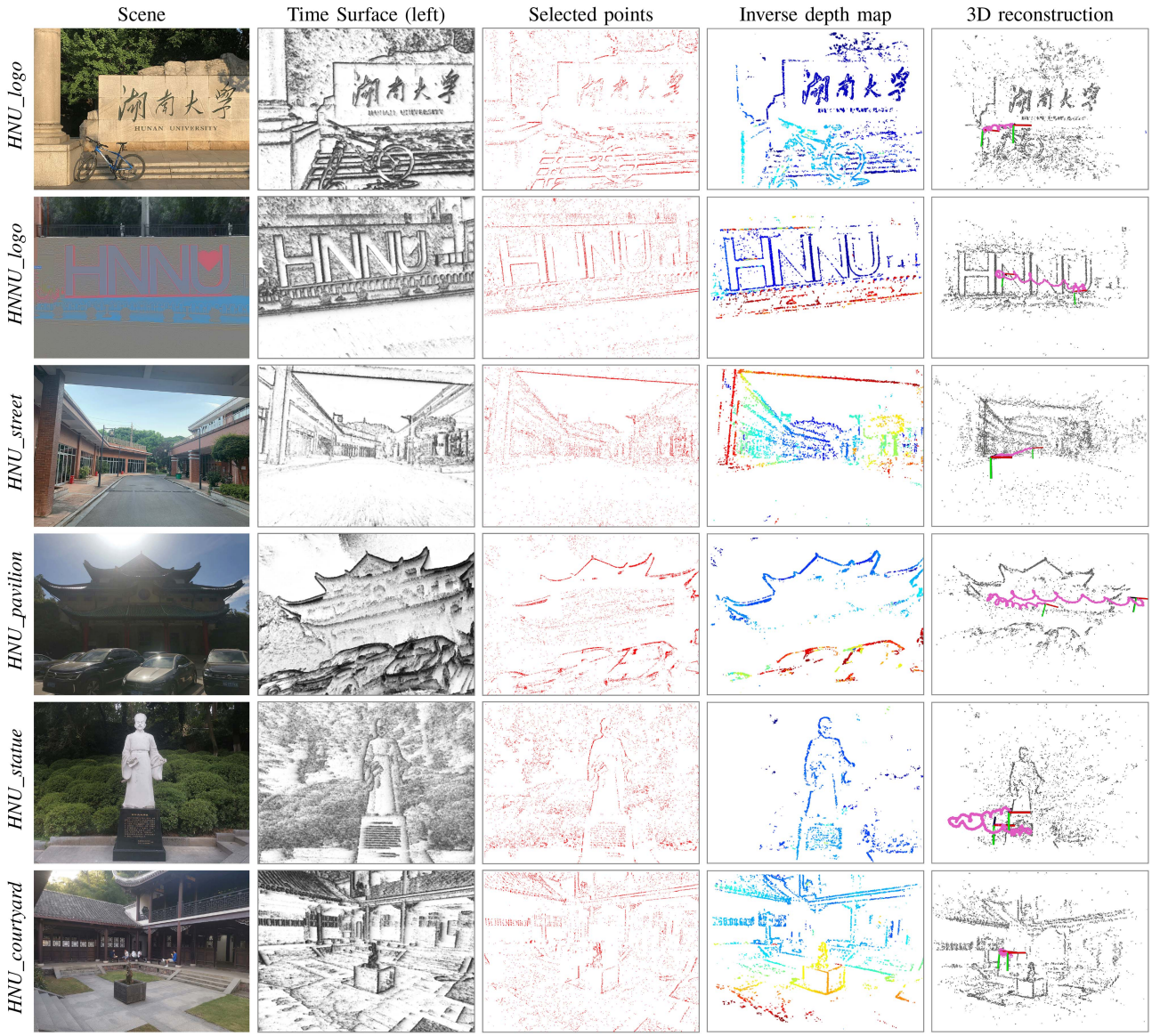



Fig. 13. *Mapping results of our system on the hnu_mapping dataset*. From left to right: scenes, time surfaces, sampled edge pixels, estimated inverse depth maps, and the resulting point clouds with the camera trajectory.

*3) Effect of the Nonlinear Refinement of Depth:* In the original ESVO pipeline [1], the disparity estimated by block matching is further refined with a successive nonlinear optimization, aiming at a subpixel accuracy. As shown in Table VIII, however, we observe that this refinement (denoted by "w/ refinement") sometimes offers only a marginal improvement in trajectory accuracy compared to the reference method while significantly increasing the runtime of the mapping operation. To guarantee a stable real-time performance at VGA resolution, we choose to omit the nonlinear refinement.

*4) Sensitivity With Respect to the Size of the Sliding Window and Necessity of Using an IMU:* To investigate how the performance changes when using different sizes of the sliding window and running the pipeline without any IMU measurements, we conduct another ablation study, as reported in Table IX. The "reference" group denotes the results obtained under the default configuration, namely, choosing a window size of 5 and using the IMU measurements. Compared to the "reference" group, the pose-estimation accuracy oscillates slightly up and down when using a window size of 8 or 12. This indicates that the choice of using different window sizes does not have a significant effect on the results. Our empirical selected window size aims to strike a balance between efficiency and accuracy. Second, it is clearly seen that the accuracy of the pose estimation drops notably when the IMU is not used, demonstrating the importance of introducing the IMU into the sensor suite.

### F. Computational Efficiency

To assess the computational efficiency, we compare the runtime of ESVO [1], ICRA'24 [42], and the proposed system using a desktop with an Intel Core i7-14700 k CPU, as shown in

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

TABLE X
COMPUTATIONAL PERFORMANCE

| **Node** (**#threads**) | **Function** | **ESVO** [1] | **ICRA'24** [42] | **Ours** |
|---|---|---|---|---|
| Pre-processing (1) | TS | 27 (∼70k) | **3.8** (∼70k) | **3.8** (∼70k) |
| | AA | - | 8 (∼70k) | **2.1** (∼70k) |
| Tracking (2) | Spatio-temporal registration | 8 (∼2k) | 7 (∼2k) | **5** (∼2k) |
| Mapping (4) | Static stereo | 46 (∼10k) | 33 (∼3.6k) | **8** (∼3.6k) |
| | Depth optimization | 78 (∼4.5k) | 13 (∼1.9k) | - |
| | Static-stereo fusion | 16.4 (∼140k) | 5.4 (∼15k) | **5.3** (∼14k) |
| | Temporal stereo | - | **4** (∼0.4k) | **4** (∼0.4k) |
| | Temporal-stereo fusion | - | **0.7** (∼2k) | **0.7** (∼2k) |
| | Regularization [optional] | 253 (∼25k) | - | - |
| | Others | **4** | 15.7 | 14.5 |
| | Subtotal [w/ optional] | 144.4 [397.4] | 71.8 | **32.5** |
| Back end (1) | Optimization | - | - | 5 |

The numbers in parentheses indicate the approximate number of event points used by each function. [Time: ms].

Table X. The tests are performed on the *DSEC* dataset. All three systems are implemented using hyperthreading technology, and the number of threads occupied by each node is declared inside the parentheses following the node name. The average runtime of each function is profiled, and the numbers in parenthesis (hyperparameters) denote the amount of data processed by each function. It should be noted that the runtime of our system is primarily influenced by the spatial resolution of the event cameras, proportionally to which the number of points sampled from AA is set. All other hyperparameters (e.g., the number of points used by each function) are set accordingly, aiming at best performance. As a result, the event streaming rate has minimal influence on our system's runtime.

We improve the rendering efficiency of TS and AA by code refactoring, resulting in TS generation being $8\times$ faster than in ESVO and AA generation being $4\times$ faster than in [42]. The efficiency of the front-end is also slightly improved due to better initial values provided by the back-end.

Most of the modifications are made in the mapping module because its performance is mostly relevant to the accuracy and efficiency of the entire system. The improvement in terms of mapping efficiency is due to the following: 1) The number of points required by the proposed mapping module is significantly reduced compared to ESVO because the input points are no longer raw events but precise contour points extracted by the applied AA and contour-point sampling method; 2) The proposed fast block matching speeds up the static-stereo operation; 3) We get rid of the depth refinement because we find that the nonlinear optimization does not bring notable improvements in mapping accuracy when the accurate contour points are fed as input; 4) The newly added temporal-stereo operation causes a runtime increment ($\approx 5$ ms). Nevertheless, it is worthwhile as it is the key to the improvement of the overall mapping quality. Moreover, the optional regularization operation is no longer used because it is time-consuming and has little effect on the front-end's performance. The "others" entry in Table X includes general operations in the code that cannot be named as functions, such as data transfer and recycle. The runtime of the mapping module demonstrates that our mapping module can stably operate in real time at 20 Hz on input stereo events of VGA resolution. Last but not the least, our back-end runs in another independent thread, taking 5 ms for each update of the optimization variables in the sliding window.

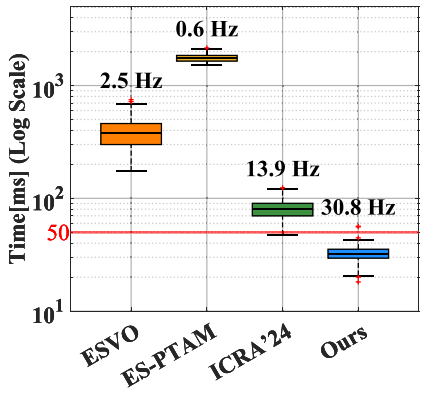



Fig. 14. *Comparison of mapping runtime.* The mean frequency of running each method's mapping module is shown above the corresponding box.

Considering that the mapping operation typically consumes the biggest portion of the computing resources in an event-based VO pipeline, we compare the mapping module's runtime performance of four open-source direct methods. To do this, we record the mapping module's runtime of every method on all VGA-resolution datasets and visualize their statistics using a box plot (see Fig. 14). It is clearly seen that only the median runtime of our method is below the 50-ms bar, ensuring real-time performance on data of VGA resolution. In addition, we assess the runtime of a feature-based method (ESIO [2]) using the same hardware. Using the recommended configuration by the authors, the average cost of the most time-consuming back-end module in ESIO is 37.34 ms. This indicates that the proposed direct method is on par with state-of-the-art feature-based methods in terms of computational efficiency; meanwhile, as shown in Sections VI-C and VI-D, our method outperforms others in accuracy.

### G. Limitations

We observe that our system does not perform well on a small number of sequences, and we attribute this to the inherent

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

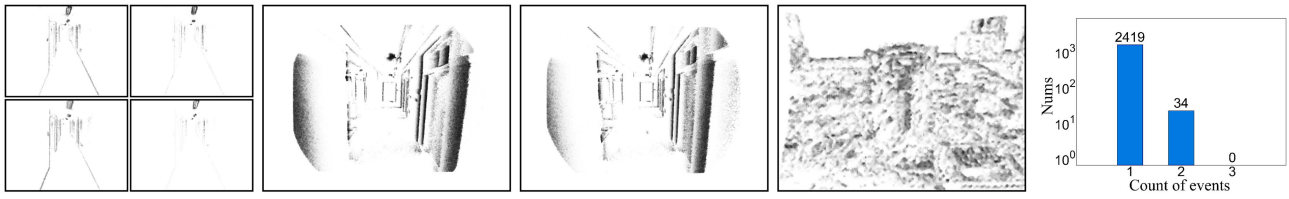



Fig. 15. *Degenerate examples and justifications.* (a) Consecutive TSs (from the left camera) selected from the *VECtor* [83] dataset, where the event edges are unstable (flickering). (b) TSs (from the left and right camera, respectively) from the *VECtor* [83] dataset, exhibiting prominent "long ramps" due to the rapid motion of edges. (c) Left: TS from the *rpg* [85] dataset. Right: the corresponding histogram made on the number of events occurred per pixel every 10 ms. (a) *corridors_dolly* [83]. (b) *school_scooter* [83]. (c) *rpg_bin* [85].

limitations of direct methods and the specific characteristics of some datasets. The first category of degenerate cases is typically witnessed in small-scale and narrow spaces, e.g., a textureless corridor [shown in Fig. 15(a) and (b)]. In this case, edge patterns either appear parallel to the moving direction of the event camera, resulting in unstable visual observations, or are so close to the event camera that the spatio-temporal profile of event data becomes significantly less distinctive, leading to data association failure in mapping and tracking. This situation exposes more the limitations of the sensor than the limitations of the proposed system. A simple cure is to introduce an additional complementary sensor, e.g., a standard camera, which easily enriches visual information. The second type of degeneration occurs when using low-resolution (e.g., 240 $\times$ 180 pixel) event cameras in cluttered environments of repeated textures. In this case, the streaming rate of events becomes lower than that using a high-resolution event camera under the same dynamics, as justified in Fig. 15(c), thus hindering the determination of contour points via the proposed AA. Furthermore, the presence of widely cluttered, repetitive textures in the scene hinders block matching, thereby worsening depth estimation.

Besides, the employed image-like representations have limitations on further (fully) exploiting the high-speed property of event cameras. We notice the existence of several alternative representations of event data [93]. For example, an early work interprets an event stream as 3-D point clouds [94] in the spatio-temporal domain, from which normal flow can be computed by fitting local planes. The computation is asynchronous and efficient (closed-form solution), and the resulting normal flow can be used for state estimation [95], leading to ultraframe-rate update rate. However, this map-free design [95] is an open-loop scheme, outperforming in speed by sacrificing accuracy. As for other representations, such as the spatio-temporal graph of events, they are typically discussed in the task of object recognition [93] but not in the context of event-based SLAM.

## VII. Conclusion

We have presented an event-based stereo visual-inertial odometry system on top of our previous work ESVO [1]. It is a direct method that solves the tracking and mapping problems in parallel by leveraging the spatio-temporal coherence in the stereo event data. The goal has been to alleviate ESVO's high computational complexity in mapping and address its degeneracy in camera pose tracking. To this end, we have additionally introduced an IMU and added several modules to the new pipeline. The newly proposed contour-point sampling method reduces the number of points required by the mapping module, significantly reducing runtime. The temporal-stereo operation acts as a complement to the static stereo, enhancing the mapping quality in terms of structure completeness and local smoothness. The tracking method is still formulated as a 3-D-2-D registration problem, and its degeneracy is addressed via the usage of IMU preintegration as motion priors and special considerations for better convergence. To maintain an accurate preintegration, a compact back-end has been proposed to suppress drift in the estimated IMU biases. Extensive experimental evaluation on five publicly available datasets of different resolutions and our own has demonstrated that our system represents the state of the art and fulfills all the claimed contributions. To the best of the authors' knowledge, the system is the first published work that achieves real-time performance using a standard CPU on event cameras of VGA pixel resolution. The software and datasets used for evaluation are open-sourced.

## Multimedia Material

Video: https://youtu.be/gmAU32Oeiv8
Code: https://github.com/NAIL-HNU/ESVO2.git

## Acknowledgment

The authors would like to thank Mr. Peiyu Chen, Dr. Zhe Liu, and Mr. Suman Ghosh for providing the raw results of [2], [63], [66], respectively, as baselines used in the evaluation. The authors would also like to thank Mr. Jinghang Li and Mr. Kaizhen Sun for the help in the data collection, and Dr. Yi Yu for proofreading.

## References

[1] Y. Zhou, G. Gallego, and S. Shen, "Event-based stereo visual odometry," *IEEE Trans. Robot.*, vol. 37, no. 5, pp. 1433–1450, Oct. 2021.
[2] P. Chen, W. Guan, and P. Lu, "ESVIO: Event-based stereo visual inertial odometry," *IEEE Robot. Autom. Lett.*, vol. 8, no. 6, pp. 3661–3668, Jun. 2023.
[3] P. Lichtsteiner, C. Posch, and T. Delbruck, "A 128$\times$ 128 120dB 15 $\mu$s latency asynchronous temporal contrast vision sensor," *IEEE J. Solid-State Circuits*, vol. 43, no. 2, pp. 566–576, Feb. 2008.
[4] G. Gallego et al., "Event-based vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 154–180, Jan. 2022.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

[5] B. Chakravarthi, A. A. Verma, K. Daniilidis, C. Fermuller, and Y. Yang, "Recent event camera innovations: A survey," in *Proc. Eur. Conf. Comput. Vis. Workshops*, 2024, *arXiv:2408.13627*.

[6] E. Mueggler, B. Huber, and D. Scaramuzza, "Event-based, 6-DOF pose tracking for high-speed maneuvers," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2014, pp. 2761–2768.

[7] X. Lagorce, C. Meyer, S.-H. Ieng, D. Filliat, and R. Benosman, "Asynchronous event-based multikernel algorithm for high-speed visual features tracking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 8, pp. 1710–1720, Aug. 2015.

[8] A. Z. Zhu, N. Atanasov, and K. Daniilidis, "Event-based feature tracking with probabilistic data association," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2017, pp. 4465–4470.

[9] G. Gallego, J. E. A. Lund, E. Mueggler, H. Rebecq, T. Delbruck, and D. Scaramuzza, "Event-based, 6-DOF camera tracking from photometric depth maps," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 10, pp. 2402–2412, Oct. 2018.

[10] G. Gallego and D. Scaramuzza, "Accurate angular velocity estimation with an event camera," *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 632–639, Apr. 2017.

[11] E. Mueggler, G. Gallego, H. Rebecq, and D. Scaramuzza, "Continuous-time visual-inertial odometry for event cameras," *IEEE Trans. Robot.*, vol. 34, no. 6, pp. 1425–1440, Dec. 2018.

[12] D. Gehrig, H. Rebecq, G. Gallego, and D. Scaramuzza, "Asynchronous, photometric feature tracking using events and frames," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 766–781.

[13] Z. Wang, T. Molloy, P. v. Goor, and R. Mahony, "Asynchronous blob tracker for event cameras," *IEEE Trans. Robot.*, vol. 40, pp. 4750–4767, 2024.

[14] J. Conradt, M. Cook, R. Berner, P. Lichtsteiner, R. J. Douglas, and T. Delbruck, "A pencil balancing robot using a pair of AER dynamic vision sensors," in *Proc. IEEE Int. Symp. Circuits Syst.*, 2009, pp. 781–784.

[15] T. Delbruck and M. Lang, "Robotic goalie with 3 ms reaction time at 4% CPU load using event-based dynamic vision sensor," *Front. Neurosci.*, vol. 7, 2013, Art. no. 223.

[16] F. Paredes-Vallés, J. J. Hagenaars, J. Dupeyroux, S. Stroobants, Y. Xu, and G. C. H. E. d. Croon, "Fully neuromorphic vision and control for autonomous drone flight," *Sci. Robot.*, vol. 9, no. 90, 2024, Art. no. eadi0591.

[17] B. Forrai, T. Miki, D. Gehrig, M. Hutter, and D. Scaramuzza, "Event-based agile object catching with a quadrupedal robot," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2023, pp. 12177–12183.

[18] Z. Wang et al., "EV-Catcher: High-speed object catching using low-latency event-based neural networks," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 8737–8744, Oct. 2022.

[19] D. Falanga, K. Kleber, and D. Scaramuzza, "Dynamic obstacle avoidance for quadrotors with event cameras," *Sci. Robot.*, vol. 5, no. 40, 2020, Art. no. eaaz9712.

[20] N. J. Sanket et al., "Evdodgenet: Deep dynamic obstacle dodging with event cameras," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2020, pp. 10651–10657.

[21] A. Bhattacharya et al., "Monocular event-based vision for obstacle avoidance with a quadrotor," in *Proc. Conf. Robot. Learn.*, 2024, *arXiv:2411.03303*.

[22] L. Pan, C. Scheerlinck, X. Yu, R. Hartley, M. Liu, and Y. Dai, "Bringing a blurry frame alive at high frame-rate with an event camera," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 6820–6829.

[23] X. Zhang and L. Yu, "Unifying motion deblurring and frame interpolation with events," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 17765–17774.

[24] Z. Zhang, A. Yezzi, and G. Gallego, "Formulating event-based image reconstruction as a linear inverse problem with deep regularization using optical flow," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8372–8389, Jul. 2023.

[25] S. Tulyakov, A. Bochicchio, D. Gehrig, S. Georgoulis, Y. Li, and D. Scaramuzza, "Time lens : Event-based frame interpolation with parametric non-linear flow and multi-scale fusion," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 17755–17764.

[26] V. Rudnev, M. Elgharib, C. Theobalt, and V. Golyanik, "EventNeRF: Neural radiance fields from a single colour event camera," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 4992–5002.

[27] A. Bhattacharya et al., "EvDNeRF: Reconstructing event data with dynamic neural radiance fields," in *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2024, pp. 5846–5855.

[28] S. Klenk, L. Koestler, D. Scaramuzza, and D. Cremers, "E-NeRF: Neural radiance fields from a moving event camera," *IEEE Robot. Autom. Lett.*, vol. 8, no. 3, pp. 1587–1594, Mar. 2023.

[29] I. Hwang, J. Kim, and Y. M. Kim, "Ev-NeRF: Event based neural radiance field," in *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2023, pp. 837–847.

[30] T. Xiong et al., "Event3DGS: Event-based 3D Gaussian splatting for high-speed robot egomotion," in *Proc. Conf. Robot. Learn.*, 2024, *arXiv:2406.02972*.

[31] H. Deguchi, M. Masuda, T. Nakabayashi, and H. Saito, "E2GS: Event enhanced Gaussian splatting," in *Proc. IEEE Int. Conf. Image Process.*, 2024, pp. 1676–1682.

[32] H. Rebecq, G. Gallego, and D. Scaramuzza, "EMVS: Event-based multi-view stereo," in *Proc. Brit. Mach. Vis. Conf.*, 2016, pp. 1–11.

[33] H. Kim, S. Leutenegger, and A. J. Davison, "Real-time 3D reconstruction and 6-DoF tracking with an event camera," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 349–364.

[34] H. Rebecq, G. Gallego, E. Mueggler, and D. Scaramuzza, "EMVS: Event-based multi-view stereo—3D reconstruction with an event camera in real-time," *Int. J. Comput. Vis.*, vol. 126, no. 12, pp. 1394–1414, 2018.

[35] H. Rebecq, T. Horstschäfer, G. Gallego, and D. Scaramuzza, "EVO: A geometric approach to event-based 6-DOF parallel tracking and mapping in real-time," *IEEE Robot. Autom. Lett.*, vol. 2, no. 2, pp. 593–600, Apr. 2017.

[36] H. Rebecq, T. Horstschaefer, and D. Scaramuzza, "Real-time visual-inertial odometry for event cameras using keyframe-based nonlinear optimization," in *Proc. Brit. Mach. Vis. Conf.*, 2017, pp. 1–12.

[37] A. R. Vidal, H. Rebecq, T. Horstschaefer, and D. Scaramuzza, "Ultimate SLAM? combining events, images, and IMU for robust visual SLAM in HDR and high speed scenarios," *IEEE Robot. Autom. Lett.*, vol. 3, no. 2, pp. 994–1001, Apr. 2018.

[38] D. G. J. Hidalgo-Carrio and D. Scaramuzza, "Learning monocular dense depth from events," in *Proc. 3D Vis.*, Nov. 2020, pp. 534–542.

[39] S. Guo and G. Gallego, "CMax-SLAM: Event-based rotational-motion bundle adjustment and SLAM system using contrast maximization," *IEEE Trans. Robot.*, vol. 40, pp. 2442–2461, 2024.

[40] S. Guo and G. Gallego, "Event-based mosaicing bundle adjustment," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 479–496.

[41] G. Taverni et al., "Front and back illuminated dynamic and active pixel vision sensors comparison," *IEEE Trans. Circuits Syst. II*, vol. 65, no. 5, pp. 677–681, May 2018.

[42] J. Niu, S. Zhong, and Y. Zhou, "IMU-aided event-based stereo visual odometry," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2024, pp. 11977–11983.

[43] G. Klein and D. Murray, "Parallel tracking and mapping for small AR workspaces," in *Proc. IEEE/ACM Int. Sym. Mixed Augmented Reality*, 2007, pp. 225–234.

[44] R. Mur-Artal, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM: A versatile and accurate monocular SLAM system," *IEEE Trans. Robot.*, vol. 31, no. 5, pp. 1147–1163, Oct. 2015.

[45] V. Vasco, A. Glover, and C. Bartolozzi, "Fast event-based Harris corner detection exploiting the advantages of event-driven cameras," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2016, pp. 4144–4149.

[46] E. Mueggler, C. Bartolozzi, and D. Scaramuzza, "Fast event-based corner detection," in *Proc. Brit. Mach. Vis. Conf.*, 2017, pp. 1–11.

[47] I. Alzugaray and M. Chli, "Asynchronous corner detection and tracking for event cameras in real time," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3177–3184, Oct. 2018.

[48] R. Li, D. Shi, Y. Zhang, K. Li, and R. Li, "FA-Harris: A fast and asynchronous corner detector for event cameras," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2019, pp. 6223–6229.

[49] C. Harris and M. Stephens, "A combined corner and edge detector," in *Proc. 4th Alvey Vis. Conf.*, 1988, vol. 15, pp. 147–151.

[50] E. Rosten and T. Drummond, "Machine learning for high-speed corner detection," in *Proc. Eur. Conf. Comput. Vis.*, 2006, pp. 430–443.

[51] I. Alzugaray and M. Chli, "ACE: An efficient asynchronous corner tracker for event cameras," in *Proc. Int. Conf. 3D Vis.*, 2018, pp. 653–661.

[52] R. I. Hartley, "In defense of the eight-point algorithm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 19, no. 6, pp. 580–593, Jun. 1997.

[53] D. Nistér, "An efficient solution to the five-point relative pose problem," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 6, pp. 756–770, Jun. 2004.

[54] S. Li, C. Xu, and M. Xie, "A robust O (n) solution to the perspective-n-point problem," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 7, pp. 1444–1450, Jul. 2012.

[55] L. Kneip, H. Li, and Y. Seo, "UPnP: An optimal O(n) solution to the absolute pose problem with universal applicability," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 127–142.

[56] A. Hadviger, I. Cvišić, I. Marković, S. Vražić, and I. Petrović, "Feature-based event stereo visual odometry," in *Proc. Eur. Conf. Mobile Robots*, 2021, pp. 1–6.


Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.

[57] J. Wang and J. D. Gammell, "Event-based stereo visual odometry with native temporal resolution via continuous-time Gaussian process regression," *IEEE Robot. Autom. Lett.*, vol. 8, no. 10, pp. 6707–6714, Oct. 2023.
[58] G. Gallego, H. Rebecq, and D. Scaramuzza, "A unifying contrast maximization framework for event cameras, with applications to motion, depth, and optical flow estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit*, 2018, pp. 3867–3876.
[59] A. Z. Zhu, N. Atanasov, and K. Daniilidis, "Event-based visual inertial odometry," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit*, 2017, pp. 5816–5824.
[60] G. Gallego, M. Gehrig, and D. Scaramuzza, "Focus is all you need: Loss functions for event-based vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 12272–12281.
[61] A. I. Mourikis and S. I. Roumeliotis, "A multi-state constraint Kalman filter for vision-aided inertial navigation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2007, pp. 3565–3572.
[62] S. Leutenegger, P. Furgale, V. Rabaud, M. Chli, K. Konolige, and R. Siegwart, "Keyframe-based visual-inertial SLAM using nonlinear optimization," in *Proc. Robot.: Sci. Syst.*, 2013, pp. 1–9.
[63] Z. Liu, D. Shi, R. Li, and S. Yang, "ESVIO: Event-based stereo visual-inertial odometry," *Sensors*, vol. 23, no. 4, 2023, Art. no. 1998.
[64] A. E. Moudni, F. Morbidi, S. Kramm, and R. Boutteau, "An event-based stereo 3D mapping and tracking pipeline for autonomous vehicles," in *Proc. IEEE Intell. Transp. Sys. Conf.*, 2023, pp. 5962–5968.
[65] S. Shiba, Y. Klose, Y. Aoki, and G. Gallego, "Secrets of event-based optical flow, depth, and ego-motion by contrast maximization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 7742–7759, Dec. 2024.
[66] S. Ghosh, V. Cavinato, and G. Gallego, "ES-PTAM: Event-based stereo parallel tracking and mapping," in *Proc. Eur. Conf. Comput. Vis. Workshops*, 2024, *arXiv:2408.15605*.
[67] S. Ghosh and G. Gallego, "Multi-event-camera depth estimation and outlier rejection by refocused events fusion," *Adv. Intell. Syst.*, vol. 4, no. 12, 2022, Art. no. 2200221.
[68] S. Shiba, Y. Aoki, and G. Gallego, "Event collapse in contrast maximization frameworks," *Sensors*, vol. 22, no. 14, pp. 1–20, 2022.
[69] U. M. Nunes, R. Benosman, and S.-H. Ieng, "Adaptive global decay process for event cameras," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 9771–9780.
[70] M. Quigley et al., "ROS: An open-source robot operating system," in *Proc. ICRA W. Open Source Softw.*, 2009, pp. 1–6.
[71] X. Lagorce, G. Orchard, F. Gallupi, B. E. Shi, and R. Benosman, "HOTS: A hierarchy of event-based time-surfaces for pattern recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 7, pp. 1346–1359, Jul. 2017.
[72] J. Manderscheid, A. Sironi, N. Bourdis, D. Migliore, and V. Lepetit, "Speed invariant time surface for learning to detect corner points with event-based cameras," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 10245–10254.
[73] A. Glover, A. Dinale, L. D. S. Rosa, S. Bamford, and C. Bartolozzi, "LuvHarris: A practical corner detector for event-cameras," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 10087–10098, Dec. 2022.
[74] Z. Wang, E. P. Simoncelli, and A. C. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. IEEE 37th Asilomar Conf. Signals, Syst. Comput.*, 2003, vol. 2, pp. 1398–1402.
[75] T. Delbruck, "Frame-free dynamic digital vision," in *Proc. Int. Symp. Secure-Life Electron.*, 2008, pp. 21–26.
[76] J. Engel, J. Stueckler, and D. Cremers, "Large-scale direct SLAM with stereo cameras," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2015, pp. 1935–1942.
[77] C. Lin, Y. Li, G. Xu, and Y. Cao, "Optimizing ZNCC calculation in binocular stereo matching," *Signal Process.: Image Commun.*, vol. 52, pp. 64–73, 2017.
[78] A. Cayley, "About the algebraic structure of the orthogonal group and the other classical groups in a field of characteristic zero or a prime characteristic," in *J. Reine Angew. Math.*, vol. 32, 1846.
[79] T. Lupton and S. Sukkarieh, "Visual-inertial-aided navigation for high-dynamic motion in built environments without initial conditions," *IEEE Trans. Robot.*, vol. 28, no. 1, pp. 61–76, Feb. 2012.
[80] C. Forster, M. Pizzoli, and D. Scaramuzza, "SVO: Fast semi-direct monocular visual odometry," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2014, pp. 15–22.
[81] T. Qin, P. Li, and S. Shen, "VINS-Mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 1004–1020, Aug. 2018.
[82] M. Gehrig, W. Aarents, D. Gehrig, and D. Scaramuzza, "DSEC: A stereo event camera dataset for driving scenarios," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4947–4954, Jul. 2021.
[83] L. Gao et al., "VECtor: A versatile event-centric benchmark for multi-sensor slam," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 8217–8224, Jul. 2022.
[84] A. Z. Zhu, D. Thakur, T. Ozaslan, B. Pfrommer, V. Kumar, and K. Daniilidis, "The multivehicle stereo event camera dataset: An event camera dataset for 3D perception," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 2032–2039, Jul. 2018.
[85] Y. Zhou, G. Gallego, H. Rebecq, L. Kneip, H. Li, and D. Scaramuzza, "Semi-dense 3D reconstruction with a stereo event camera," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 242–258.
[86] S. Klenk, J. Chui, N. Demmel, and D. Cremers, "TUM-VIE: The TUM stereo visual-inertial event dataset," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2021, pp. 8601–8608.
[87] P. Chen et al., "ECMD: An event-centric multisensory driving dataset for SLAM," *IEEE Trans. Intell. Veh.*, vol. 9, no. 1, pp. 407–416, Jan. 2024.
[88] T. Pönitz, J. Stöttinger, R. Donner, and A. Hanbury, "Efficient and distinct large scale bags of words," in *Proc. 34th Annu. Workshop Austrian Assoc. Pattern Recognit. (AAPR) WG Vis. Comput. Austrian Comput. Soc.*, 2010, pp. 139–146.
[89] K. Chaney et al., "M3ED: Multi-robot, multi-sensor, multi-environment event dataset," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, 2023, pp. 4016–4023.
[90] D. Gehrig and D. Scaramuzza, "Are high-resolution event cameras really needed?," 2022, *arXiv:2203.14672*.
[91] T. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, and D. Rus, "LIO-SAM: Tightly-coupled lidar inertial odometry via smoothing and mapping," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, 2020, pp. 5135–5142.
[92] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of RGB-D SLAM systems," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, Oct. 2012, pp. 573–580.
[93] Y. Li et al., "Graph-based asynchronous event processing for rapid object recognition," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 934–943.
[94] R. Benosman, C. Clercq, X. Lagorce, S.-H. Ieng, and C. Bartolozzi, "Event-based visual flow," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 2, pp. 407–417, Feb. 2014.
[95] X. Lu, Y. Zhou, J. Niu, S. Zhong, and S. Shen, "Event-based visual inertial velometer," in *Proc. Robot.: Sci. Syst.*, 2024, pp. 1–11.



**Junkai Niu** received the master's degree in aerospace science and technology from the University of Electronic Science and Technology of China, Chengdu, China, in 2023. He is currently working toward the Ph.D. degree in control science and engineering with the Neuromorphic Automation and Intelligence Lab, School of Robotics, Hunan University, Changsha, China.

His research interests primarily focus on event-based visual odometry/simultaneous localization and mapping.



**Sheng Zhong** received the B.Eng. degree in robotics engineering from Zhejiang University, Hangzhou, China, in 2023. He is currently working toward the Ph.D. degree with the Neuromorphic Automation and Intelligence Lab, School of Robotics, Hunan University, Changsha, China.

His research interests primarily focus on event-based visual odometry/simultaneous localization and mapping.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.



**Xiuyuan Lu** received the B.Eng. degree in computer science in 2020 from the Hong Kong University of Science and Technology (HKUST), Hong Kong, where he is currently working toward the Ph.D. degree in electronic and computer engineering.

His research interests include event-based vision and visual odometry/simultaneous localization and mapping.



**Shaojie Shen** received the B.Eng. degree in electronic engineering from the Hong Kong University of Science and Technology (HKUST), in 2009, the M.S. degree in robotics and the Ph.D. degree in electrical and systems engineering from the University of Pennsylvania, Philadelphia, PA, USA, in 2011 and 2014, respectively.

He is currently an Associate Professor with the Department of Electronic and Computer Engineering and the founding Director of the HKUST-DJI Joint Innovation Laboratory (HDJI Lab), HKUST. His research interests include the areas of robotics and unmanned aerial vehicles, with focus on state estimation, sensor fusion, localization and mapping, and autonomous navigation in complex environments.

Dr. Shen and his research team were the recipients of the 2023 IEEE T-RO King-Sun Fu Memorial Best Paper Award and the 2023 IEEE RA-L Best Paper Award. He was also the recipient of the Honorable Mention status for the IEEE T-RO Best Paper Award in 2018 and 2020, and the Best Student Paper Award in IROS 2018. In addition, was recognized as the AI 2000 Most Influential Scholar Award Honorable Mention in 2020 and consecutively from 2021–2024. He is currently serving as a Senior Editor for ICRA 2024–2026, and was an Associate Editor for IJRR 2023–2024.



**Guillermo Gallego** received the Ph.D. degree in electrical and computer engineering from the Georgia Institute of Technology, Atlanta, GA, USA, in 2011.

From 2011 to 2014, he was a Marie Curie Researcher with Universidad Politecnica de Madrid, Spain, and from 2014 to 2019, he was a Postdoctoral Researcher with the Robotics and Perception Group, University of Zurich and ETH Zurich, Switzerland. He is currently a Full Professor with the Department of Electrical Engineering and Computer Science, Technische Universität Berlin, Berlin, Germany, and with the Einstein Center Digital Future, Berlin. He is also a Principal Investigator with the Science of Intelligence Excellence Cluster and the Robotics Institute Germany, Berlin. His research interests include robotics, computer vision, signal processing, optimization, and geometry.

Dr. Gallego was the recipient of a Fulbright Scholarship. He serves as Associate Editor for IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, and for IEEE ROBOTICS AND AUTOMATION LETTERS, the *International Journal of Robotics Researc*h, and as Guest Editor of the IEEE TRANSACTIONS ON ROBOTICS special collection on event-based vision for robotics.



**Yi Zhou** (Member, IEEE) received the Ph.D. degree in engineering and computer science from the Australian National University, Canberra, ACT, Australia, in 2018.

From 2017 to 2018, he was a Visiting Scholar with ETH Zurich. From 2019 to 2021, he was a postdoc Research Fellow with the HKUST and DJI Innovation Joint Lab, where he proposed the world's first open-source event-based stereo visual odometry (ESVO) system. He is currently a Professor with Hunan University, Changsha, China, where he directed the Neuromorphic Automation and Intelligence Lab (NAIL). His research interests include visual odometry / simultaneous localization and mapping, geometry problems in computer vision, and dynamic vision sensors.

Prof. Zhou was the recipient of the NCCR Fellowship Award by the Swiss National Science Foundation for his research on neuromorphic event-based 3-D vision.

Authorized licensed use limited to: Technion Israel Institute of Technology. Downloaded on September 11,2026 at 02:01:39 UTC from IEEE Xplore. Restrictions apply.